7902 Markham Firebirds — Manufacturing Curriculum

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Date Updated: June 2024
Date Created: June 30, 2024

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1.1 Engineering Design Process

Engineering design is a systematic and creative problem-solving process used by engineers and designers to develop innovative solutions to various situations and challenges. It is a structured approach that helps individuals or teams identify, conceptualize, plan, create, and improve upon solutions. The process can be broken down into several key steps:

1. **ASK**: This initial stage involves identifying and defining the problem. Engineers start by asking questions and conducting research to understand the problem's root cause, requirements, constraints, and objectives. They gather data and information related to the issue they are trying to address.

2. **IMAGINE**: Once the problem is well understood, engineers use their creativity and knowledge to generate a wide range of possible solutions. During this stage, brainstorming sessions, sketches, simulations, and other tools are used to explore various ideas and concepts. The goal is to come up with as many potential solutions as possible without considering common limitations such as feasibility, cost, resources just to name a few.

3. **PLAN**: In this phase, the most promising ideas from the IMAGINE stage are evaluated and refined. Engineers consider factors such as feasibility, cost, resources, safety, and environmental impact. Detailed plans are developed, including the selection of materials, manufacturing processes, and timelines.

4. **CREATE**: With a solid plan in place, engineers proceed to build a prototype or actual

product. This step involves transforming the chosen design from the planning stage into a tangible form. Depending on the complexity of the project, multiple iterations may be required to achieve the desired outcome.

5. **TEST**: Once the creation is complete, engineers subject the solution to rigorous testing and analysis. Testing helps to verify if the design meets the specified requirements and performs as expected. Engineers gather data and feedback to identify any shortcomings or areas for improvement.

6. **IMPROVE**: After analyzing the test results, engineers assess the performance of the solution and compare it to the initial objectives. If necessary, they make modifications to enhance the design, address any issues, or optimize its performance. This process of continuous improvement is crucial for refining the solution and achieving better results.

7. **DISCUSS AND REPEAT**: Engineering design is rarely a solitary endeavor. Collaboration and communication within the team are vital throughout the entire process. Engineers discuss the results, share ideas, and seek feedback from colleagues to gain different perspectives. If further improvements are needed, the process is repeated, incorporating the insights gained from testing and discussions.

By following this systematic approach, engineers can develop effective and innovative solutions to a wide range of problems across various fields, including mechanical, electrical, civil, aerospace, software, and more. It ensures that the final product or system meets the intended objectives and performs optimally while adhering to safety and ethical considerations. Additionally, the iterative nature of the engineering design process allows for continuous enhancement and adaptation in response to changing requirements and advancements in technology.

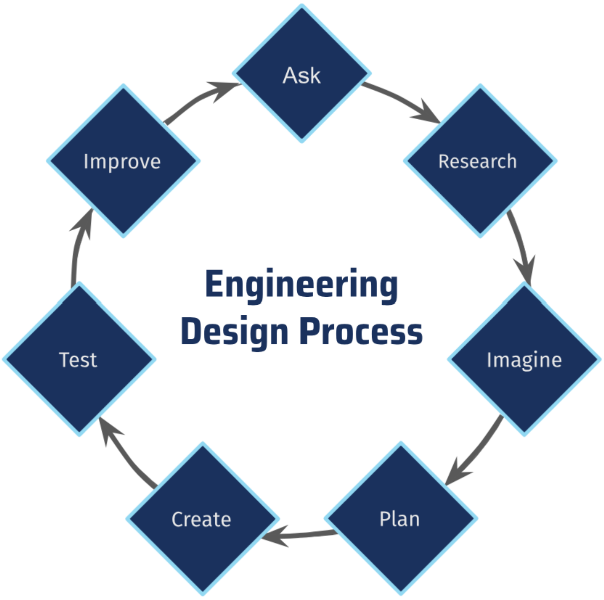

Figure 1: Engineering Design Process

1.2 FRC Design Process

The FRC Design Process is a structured approach used by teams participating in the FIRST Robotics Competition to design, build, and improve their robot. It consists of several stages that guide teams through the process of creating a competitive robot to tackle the specific challenges presented in each year's competition. Let's break down each stage:

1. **Rule Analysis**: In this stage, the team thoroughly analyses the rules and requirements of the current year's FRC game. They study the game manual and understand the tasks the robot needs to perform, the scoring mechanisms, and any constraints or limitations imposed by the rules.

2. **Strategic Design**: Similar to the "Ask" stage, the team strategies and decides on the overall approach for the robot. They determine what the robot should do to score points effectively, considering the game objectives and the team's overall strategy for winning the competition. They also assess their team's capabilities and resources to make realistic decisions.

3. **Conceptualize**: This stage involves brainstorming and generating concepts for the robot design. The team decides on the specific functions and tasks the robot should be able to perform to meet the game's objectives. They also consider the various subsystems needed to accomplish these functions. During this phase, the team creates rough sketches and drawings to visualize the robot's layout, subsystems, and mechanisms. These concept sketches serve as the initial blueprint for the robot's design.

4. **Plan**: In this phase, the team takes the concept sketches and creates detailed plans for the robot. This includes using Computer-Aided Design (CAD) software to model the robot, specifying the materials required, creating a build schedule, and ensuring that the robot's functionality aligns with the strategic goals.

5. **Build**: The build stage is when the team brings the design to life. They manufacture robot parts, order necessary materials, assemble components, and program the robot's control systems. It is the phase where the physical robot starts taking shape.

6. **Test**: Testing is a critical step in the FRC Design Process. The team performs both pre-competition testing and drive practice to evaluate the robot's performance, identify any issues or inefficiencies, and make necessary adjustments. During the actual competition, the team continues to test and improve the robot based on its performance in real-game scenarios. They observe how the robot operates in the competition field and gather valuable data to further optimize its performance.

7. **Improve**: After each competition, the team assesses the robot's performance and identifies areas that require improvement. They use this feedback to enhance the robot's design and functionality, making it more effective for subsequent competitions.

The FRC Design Process is iterative, meaning that the team repeats certain stages, such as "Test" and "Improve," multiple times to continuously enhance the robot's performance and refine their

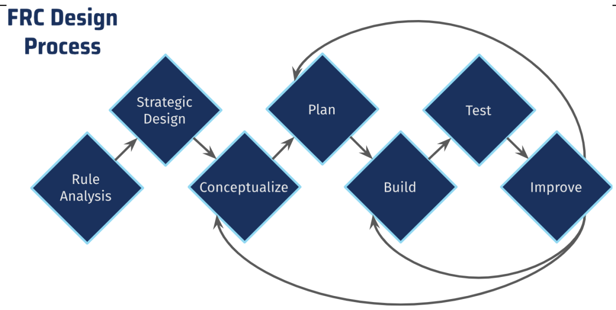

Figure 2: FRC Design Process

strategies. This process allows FRC teams to approach the competition systematically, improving their chances of success by creating a well-designed and robust robot.

1.3 Difference Between Industrial Design and Engineering Design

There is a common misconception between what industrial design and engineering design and what they mean and represent.

1.3.1 Industrial Design

Industrial design is a multifaceted discipline that blends creativity, aesthetics, psychology, engineering, and business savvy. The primary goal is to craft products that are visually appealing, function flawlessly, and enhance users' lives. Achieving successful industrial design results in higher product adoption rates, increased brand loyalty, and overall customer satisfaction.

In the pursuit of well-designed products, industrial designers carefully consider the relationship between components, ensuring harmonious visuals while optimizing usability. Thoughtful material selection and texture choices contribute to the product's tactile appeal and purposeful functionality. Colours are strategically used to evoke emotions and align with the brand's identity, fostering a memorable and cohesive user experience.

Additionally, industrial designers focus on ergonomics, creating products that offer comfort and efficiency during use. Collaboration with engineering and manufacturing teams ensures that the designs can be brought to life efficiently and within budget. By delivering a compelling user experience that resonates with consumers, industrial design plays a crucial role in establishing brand loyalty and driving long-term success in the market.

1.3.2 Engineering Design

The Engineering Design Process is a problem-solving approach utilized by engineers to tackle challenges effectively. It comprises seven stages: ask, research, imagine, plan, create, test, and improve. This versatile process can be applied in various situations to identify and address problems efficiently.

Adapted to fit the FRC design process, the Engineering Design Process takes on a more linear structure. While some stages remain similar, others may differ to suit the specific requirements of the FIRST Robotics Competition. Despite these adjustments, the fundamental principles of problem-solving and continuous improvement remain at the core of both approaches.

Overall, whether applied in standard engineering projects or adapted for FRC, the Engineering Design Process empowers engineers and teams to develop innovative solutions and overcome challenges successfully. Its adaptability makes it a valuable framework for problem-solving across a wide range of contexts.

Design Conclusion

Industrial design is a dynamic discipline that merges creativity, aesthetics, engineering, and business acumen to create visually appealing and functional products that enrich users' lives. By carefully considering the relationship between components, materials, and colours, industrial designers craft products that resonate with consumers and foster brand loyalty. Moreover, emphasis on ergonomics ensures that products offer comfort and efficiency during use. Collaborating with engineering and manufacturing teams ensures the successful realization of designs, leading to a compelling user experience and long-term market success.

The Engineering Design Process, a problem-solving approach employed by engineers, offers a versatile and systematic framework for tackling challenges effectively. While adapted to fit the unique requirements of the FRC design process, its fundamental principles of problem-solving and continuous improvement remain unchanged. Whether applied in standard engineering projects or tailored for FRC, the Engineering Design Process empowers teams to develop innovative solutions and navigate obstacles with confidence. Its flexibility makes it a valuable tool for problem-solving across various contexts, contributing to successful outcomes in engineering and robotics endeavors.

1.4 Problem Solving

Problem-solving is a crucial aspect of the FRC design process, requiring continuous application at every stage. During rule analysis, teams must think critically about how each rule impacts game play and strategize how to leverage them to their advantage while anticipating competitors' strategies.

Strategic design involves problem-solving to generate efficient and quick robot ideas that can accomplish the tasks effectively. This leads to devising subsystem designs that work cohesively and fit onto the robot, solving challenges related to task completion and integration.

The planning stage introduces new problem-solving opportunities, like sourcing materials, addressing building complexities, and ensuring seamless subsystem attachment. Challenges during building arise, such as materials not cooperating, game pieces not reacting as intended, and insufficient tool knowledge.

Testing the prototype/subsystem highlights potential issues such as inadequate shooting distance, misaligned angles, or space constraints. Some problems can be addressed during testing, while others may require revisiting earlier stages to refine designs and solutions.

Throughout the FRC design process, problem-solving skills are indispensable, allowing teams to identify, address, and overcome obstacles effectively, ultimately leading to the development of a successful and competitive robot. Continuous evaluation and adaptation contribute to optimal performance and improvements at each stage of the process.

Problem Solving Conclusion

Problem-solving is a cornerstone of the FRC design process, guiding teams at every stage. Analyzing rules requires critical thinking to strategize effectively and anticipate competitors' moves. Strategic design yields efficient robot ideas, integrating subsystems to tackle challenges seamlessly.

The planning stage presents new problem-solving opportunities, from sourcing materials to addressing building complexities. Challenges arise during building, demanding creative solutions for uncooperative materials and unpredictable game pieces. Testing uncovers potential issues, prompting further refinement.

Throughout the process, problem-solving skills are crucial, empowering teams to overcome obstacles and create successful, competitive robots. Continuous evaluation and adaptation lead to optimal performance and continual improvement. Embracing problem-solving equips teams for triumph in the dynamic world of FRC robotics

2 FRC Game Structure

In a FRC game, different elements contribute to the overall game play experience, including autonomous, tele-op, end game, ranking points, and scoring periods. In the upcoming sections we will talk about the different elements that work to contribute to the overall game play and strategy.

2.1 Alliance

In a match there are 2 alliances, a Red alliance and a Blue alliance. On each alliance there are 3 teams, so overall in each math there are 6 teams playing at one time. Cooperation with your alliance members can very well determine your success in the match. Discussing strategy with your alliance members help to determine a strategy that works with your teams strength and help make up for weaknesses.

2.2 Autonomous

During the autonomous period, robots are guided based on pre-built code, allowing them to perform tasks without direct driver input. In certain games, such as FRC 2019 Deep Space, drivers may be able to use cameras to monitor and control the robot. However, for the most part, robots rely solely on their programmed instructions to carry out specific actions during the autonomous period as the drive team is required to stand behind the line in the driver station leading to no control of the robot. Usually there will be game pieces for the game, whether it be a cube, ring, ball, or another shape you usually are allow to pre-load a game piece to start autonomous period with. This again means that this phase needs precise planning, fine tuning and problem-solving skills to develop efficient and effective routines that can score points, gain advantages in the game, and cooperate with your alliance members to score the most points.

2.3 Tele-Op

In contrast, the tele-op period involves direct human control of the robots. Drivers skillfully manoeuvre their robots on the field, responding to real-time challenges and dynamically adapting to the game's evolving dynamics. Effective problem-solving during tele-op is essential for making strategic decisions, coordinating with teammates, and outmanoeuvring opponents to maximise scoring opportunities.

2.4 End Game

During the end game, teams face critical decision-making moments as they aim to secure additional points or execute game-specific actions to gain advantages in the final moments of the match. Problem-solving skills come into play as teams strategize and prioritise their end-game objectives, aiming for the most impactful outcomes.

2.5 Ranking Points

Ranking points play a vital role in FRC competitions, influencing team standings and playoff eligibility. Problem-solving becomes essential in determining the most effective strategies for earning ranking points, such as achieving specific game objectives or collaborating with alliance partners to maximise cumulative scores.

2.6 Scoring

Scoring is defined as any action that earns the alliance game points. Game points don't count towards the ranking point. The winner of a game is determined by the game points. Any alliance that wins a game is awarded 2 ranking points, if they tie then both alliances earn 1 ranking point, and if they lose they are not awarded any ranking points.

Ultimately, the scoring system provides a clear measure of success, and teams must continuously analyse and refine their strategies to optimise scoring opportunities throughout the game. Problem-solving skills are instrumental in identifying weaknesses, adjusting tactics, and leveraging strengths to achieve higher scores and secure victories in the FRC game.

Game Structure Conclusion

In conclusion, problem-solving is a fundamental and integral part of the FRC design process, playing a critical role at every stage of robot development. From rule analysis and strategic design to planning, building, testing, and adapting, teams must continuously apply problem-solving skills to overcome challenges and create efficient and competitive robots.

The FRC game structure presents various opportunities for problem-solving, starting with the autonomous period, where teams must develop precise and effective routines to score points and gain advantages. During the tele-op phase, problem-solving becomes essential for drivers to make strategic decisions, coordinate with teammates, and outmanoeuvre opponents to maximise scoring opportunities in real-time.

The end game presents critical decision-making moments, and problem-solving skills are instrumental in strategizing and prioritising end-game objectives to secure additional points and advantages in the final moments of the match. Furthermore, understanding the ranking point system and scoring dynamics allows teams to devise effective strategies to earn ranking points, ultimately influencing team standings and playoff eligibility.

Throughout the FRC competition, problem-solving skills are indispensable for identifying weaknesses, adjusting tactics, and leveraging strengths to achieve higher scores and secure victories. By continuously analysing and refining their strategies, teams can optimise scoring opportunities and position themselves for competitive success in the FRC game. Emphasising problem-solving at each stage of the design process leads to the development of successful and high-performing robots, contributing to a rewarding and challenging FRC experience for all teams involved.

3 Game Manual

Before embarking on robot design, thoroughly reviewing the game manual is of utmost importance. Overlooking this step can lead to frustration and setbacks later in the process. The game manual contains crucial rules and guidelines specific to the year's competition, and failure to adhere to these regulations can result in violations, including exceeding build limits or other crucial specifications that vary each year.

To ensure comprehensive rule compliance and effective design, a practical approach is to divide the team into three groups, each assigned to investigate three specific topics. This division of labour allows for in-depth examination of the rules, as different teams focus on distinct aspects. Through this collective effort, teams can gain a comprehensive understanding of the game manual's requirements, enabling them to craft a well-informed and compliant robot design.

By reading the game manual, teams safeguard their design process from potential pitfalls and complications arising from rule violations. Understanding the game's specific constraints empowers teams to make strategic decisions during the design phase, optimising their robot's capabilities and positioning themselves for a competitive edge in the FRC competition.

3.1 Group 1 - Overview

This group is tasked with studying the general layout of the game, focusing on various key aspects and taking comprehensive notes. Areas to investigate include, but are not limited to:

1. Different areas of the playing field (e.g., baseline, scoring zones, climb locations): Analysing the purpose and significance of each area and how they contribute to the game's overall dynamics.

2. Location of field pieces: Identifying the positioning of critical elements on the field, such as obstacles or game pieces, that can influence gameplay strategies.

3. On-field game pieces: Examining the characteristics and properties of game pieces used during the competition, understanding how they interact with the robot and contribute to scoring opportunities.

4. Loading zones: Investigating specific regions on the field designated for loading game pieces onto the robot during the match.

5. Preloading: Understanding the rules and limitations regarding preloading game pieces onto the robot before the start of the match.

6. Scoring zones: Delving into how and where game pieces are scored to earn points during the competition, as well as the corresponding scoring mechanisms.

7. Starting location: Determining the robot's initial position on the field and how it impacts the team's overall strategy and game plan.

By thoroughly exploring these areas and documenting their findings, the group gains a comprehensive understanding of the game's layout and dynamics. This knowledge is essential for strategizing, designing, and optimising the robot's capabilities to excel in the FRC competition.

3.2 Group 1 - Actions

This group is responsible for analyzing all the possible actions a robot can undertake during each period of the game: autonomous, tele-op, and end game. Their task includes but is not limited to:

1. Autonomous period: Studying the various actions the robot can execute without driver input during the initial stage of the match. This involves understanding pre-programmed routines, movement sequences, and interactions with game elements to gain an early advantage in scoring.

2. Tele-op period: Examining the wide array of actions the robot can perform under driver control during the majority of the match. This includes tasks such as collecting game pieces, delivering them to scoring zones, playing defense, and cooperating with alliance partners for strategic gameplay.

3. End game: Investigating the unique actions available to the robot during the final moments of the match, often offering significant scoring opportunities or crucial game-changing manoeuvres. This may involve climbing, hanging, or engaging in other actions specific to the game's end game mechanics.

By thoroughly analyzing and documenting all the potential robot actions across these game periods, this group gains valuable insights into the range of strategies and possibilities available to their team. Understanding the robot's capabilities and limitations during each phase empowers teams to create effective game plans, devise winning strategies, and optimize their overall performance in the FRC competition.

3.2.1 Chronologically Dependent Actions

When examining the chronological dependent actions, it is essential to note the specific game periods during which each action can be executed—autonomous, tele-op, or endgame. Creating a comprehensive list of these actions and categorizing them based on their timing is a crucial step in understanding the robot's capabilities and strategizing effectively.

By organizing the actions into different categories according to their respective game periods, teams gain a clear overview of the robot's available options at each stage of the match. This enables them to plan and priorities their actions strategically, optimising scoring opportunities and maximizing their robot's contribution to the team's overall game plan.

Categorizing the actions also facilitates the process of devising autonomous routines, tele-op strategies, and endgame manoeuvres. It allows teams to focus on developing specialized skills and efficient sequences for each game period, ensuring that the robot operates seamlessly and effectively throughout the entire match.

In conclusion, by taking meticulous notes of the chronological dependent actions and sorting them into relevant categories based on their timing, teams can enhance their understanding of the game dynamics and tailor their robot's capabilities to perform optimally in each phase of the competition. This systematic approach contributes to a well-coordinated and competitive performance in the FRC tournament.

3.2.2 Action List

When creating the action list, it is crucial for the group to consider not only the point-earning actions but also the non-point-earning actions. The comprehensive list should encompass various categories, including:

1. **Non-scoring Movement Techniques**: This category focuses on actions related to the

robot's movement that do not directly score points. It includes driving manoeuvres, strategic positioning, and movements to gain a tactical advantage on the field.

2. **Simple Scoring Actions**: Actions falling under this category refer to relatively straightforward manoeuvres that can directly score points for the team. For example, crossing the initiation line during autonomous can earn specific points.

3. **Game Piece Actions**: This category involves actions related to handling game pieces during the match. It includes tasks such as picking up game pieces from the ground or alliance stations, as well as passing game pieces to alliance robots for cooperative gameplay.

4. **Assisting Actions**: Actions in this category focus on supporting and assisting ally robots during the match. This may include actions like providing a strategic block to deter opponents, clearing paths, or executing moves to set up advantageous positions for alliance partners.

By encompassing all these aspects in the action list, the group gains a comprehensive understanding of the robot's capabilities and strategic possibilities throughout the game. This information is invaluable for devising well-coordinated game plans, optimising scoring opportunities, and enhancing teamwork and collaboration with alliance partners during the FRC competition. A thorough and inclusive action list enables teams to make informed decisions, adapt to dynamic match situations, and perform competitively at the highest level.

3.2.3 Maximum Points

When analyzing each action with an associated point value, the group's task is to calculate the theoretical maximum points a single bot can earn in a single match. This calculation should consider various factors, including the number of game pieces a robot can carry at once, the available scoring locations on the field, and the total number of game pieces available during the match.

Taking into account these variables allows the group to assess the robot's scoring potential and strategize how to optimize point accumulation. By identifying the maximum points attainable for each action, teams can priorities high-scoring tasks and develop effective gameplay strategies.

It is essential to be mindful of actions with potentially "infinite" possibilities, where the number of points a robot can earn is practically unlimited. In such cases, these actions should be flagged for further analysis and revisited later in the assessment process. Understanding the full extent of these "infinite" scoring opportunities is crucial in fine-tuning the robot's overall strategy and identifying areas for potential domination on the field.

By conducting this comprehensive analysis, teams can gain valuable insights into the robot's scoring potential and strategize for optimal point accumulation. This data-driven approach contributes to the development of a competitive and high-performing robot capable of maximizing its scoring capabilities in the FRC competition.

3.2.4 Ranking Points

In addition to actions that directly earn points, alliances in the competition can also pursue specific actions or "awards" that grant them extra ranking points. An example from the 2020 game involved earning a ranking point by achieving certain milestones during the end game period. This included scoring at least 65 points in the end game or activating the 3rd stage for the shield generators.

These additional ranking points incentive alliances to focus on specific game objectives that may significantly impact their overall ranking in the competition. By pursuing these awards, alliances can strategically optimize their scoring potential and increase their chances of ranking higher on the leader board.

Understanding the criteria for earning these bonus ranking points is crucial in developing a well-rounded and strategic game plan. Teams can assess the feasibility of achieving these objectives while also balancing their efforts on actions that directly contribute to point accumulation during the match.

By prioritizing actions that earn ranking points, alliances can not only boost their position in the standings but also enhance their chances of being selected for playoffs and forming successful alliances with other top-performing teams. Strategically pursuing these awards contributes to a more competitive and successful FRC campaign.

3.2.5 Bonus Points

In certain situations, points can be earned by repeating specific actions or a series of actions during the match. To maximize scoring potential, it is essential to identify and record every possible way your robot can achieve these bonus points.

Recording all viable strategies for earning bonus points allows the team to explore various scenarios and assess the feasibility of executing them effectively during the competition. By meticulously documenting these opportunities, teams can develop a comprehensive playbook of scoring strategies and choose the most optimal approach based on match dynamics and alliance strategies.

Moreover, understanding all the ways to earn bonus points facilitates adaptive gameplay during the match. Teams can react quickly to changing conditions and capitalize on emerging opportunities to score additional points. This flexibility in executing different scoring strategies enhances the team's competitiveness and ability to secure crucial ranking points during the FRC competition.

By recording and analyzing all viable methods to achieve bonus points, teams can develop a strategic advantage, increase their scoring potential, and elevate their overall performance in the game. A well-documented and versatile approach to scoring contributes to the team's success, leading to higher rankings and more opportunities for playoff advancement and alliance selection.

3.3 Restrictions

As highlighted in Section 5 - Game Manual, building the robot involves adhering to specific rules and restrictions. Robot restrictions encompass various limitations that teams must follow during the construction process. In previous games, these restrictions have encompassed aspects such as bumper regulations, extension limitations, requirements to keep the robot in one piece, and other specific robot-related constraints.

The group assigned to this section should diligently take notes on what constitutes a violation of these restrictions. Understanding the precise boundaries and limitations ensures that the robot remains compliant with the rules and avoids penalties or disqualifications during the competition.

By meticulously documenting potential violations, teams can effectively analyze their design choices, ensuring that their robot adheres to all rules and restrictions while still maximizing its capabilities. Adhering to these restrictions allows teams to compete on a level playing field, promoting fair competition and fostering a spirit of innovation and creativity within the constraints of the FRC competition.

Game Manual Conclusion

Before delving into the robot design process, conducting a meticulous review of the game manual is paramount. The game manual serves as a comprehensive guide containing essential rules and guidelines specific to the year's competition. Ignoring this crucial step can lead to potential rule violations, penalties, or disqualifications during the competition, causing setbacks and frustrations for the team. Understanding the intricacies and constraints of the game empowers teams to make informed decisions during the design phase, ensuring that their efforts are well-directed and resources are optimally allocated.

To comprehensively examine the game manual, teams can divide themselves into smaller groups, each assigned to investigate specific topics. This approach allows for a deeper understanding of the rules, as different teams can focus on distinct aspects of the game manual. One group can concentrate on studying the general layout of the game, understanding different areas of the playing field, locations of field pieces, loading zones, scoring zones, starting positions, and more. Another group can analyze the robot's possible actions during different game periods, including autonomous, tele-op, and endgame. By understanding the robot's capabilities and limitations during each phase of the match, teams can develop effective game play strategies and optimise overall performance.

Moreover, teams should pay special attention to actions that earn points, bonus ranking points, and ways to achieve bonus points through repeated actions. Calculating the theoretical maximum points that a robot can earn in a single match helps identify high-scoring tasks and strategic actions that can significantly impact overall match performance. Additionally, understanding actions that grant bonus ranking points incentives teams to focus on specific game objectives, potentially elevating their ranking on the leader board and increasing opportunities for playoff advancement and alliance selection.

Lastly, teams must carefully review and adhere to any restrictions outlined in the game manual. These restrictions encompass various limitations that teams must follow during the robot's construction process. Complying with these rules ensures fair competition and prevents penalties or disqualifications that could hinder the team's performance. By thoroughly studying the game manual and considering all aspects of the competition, teams can strategize effectively, optimise their robot's capabilities, and position themselves for competitive success in the FRC competition.

4 Drawings

In this lesson, participants will have the opportunity to collaborate closely with the CAD subdivision, gaining hands-on experience in interpreting their technical drawings and transforming those designs into the actual components required for a project. The lesson focuses on two primary types of drawings provided by the CAD subdivision: isometric and orthographic. Through practical exercises and guidance, participants will learn to distinguish between these two schematic drawing styles and develop the skills needed to accurately interpret and translate them into tangible products.

The CAD subdivision's drawings serve as essential blueprints for constructing various components in engineering and design projects. Understanding the difference between isometric and orthographic drawings is crucial, as each offers unique perspectives and information about the spatial arrangement and dimensions of the objects being depicted. Participants will be introduced to the concepts of isometric projection, which provides a three-dimensional view of an object, and orthographic projection, which presents multiple two-dimensional views from different angles.

Throughout the lesson, students will gain proficiency in interpreting and comprehending the intricate details conveyed in these technical drawings. By mastering this skill, they will be able to accurately recreate the intended components, ensuring precision and efficiency in the manufacturing or construction process. The ability to work seamlessly with the CAD subdivision's drawings and implement the designs effectively will be a valuable asset for any engineering or design professional.

4.1 Isometric Drawing

An isometric drawing is a method used to represent a 3D object in a 2D image without distortion. It offers a clear, three-dimensional visualisation by maintaining equal scales along all three axes. This technique is commonly used in engineering, architecture, and design to provide a realistic and easily understandable portrayal of objects and structures. Isometric drawings enable viewers to grasp the spatial relationships and proportions of the depicted object accurately, facilitating effective communication and problem-solving in various industries.

4.1.1 Construction of an Isometric Drawing

When creating an isometric drawing, depth perception does not need to be considered. To construct an isometric drawing, vertical lines should be drawn perpendicular to the bottom of the paper or connecting two dots vertically on isometric paper. For horizontal lines, a 30-degree angle to the bottom of the paper or connecting two dots diagonally on isometric paper is used. By following these guidelines, the isometric drawing accurately represents the object's three-dimensional appearance while maintaining a visually balanced and distortion-free representation of the object. Isometric drawings are valuable tools in various fields, aiding in visual communication and efficient design processes.

The most best way to learn isometric drawings is through hands-on practice. Simply grab a pencil and attempting to create an isometric drawing on your own. To get started, you can find numerous isometric paper templates with dots online or use plain white paper. Select a few simple objects like a book or a pencil case from your surroundings and try sketching their isometric representations. This practical approach allows you to grasp the principles of isometric drawing and improve your skills through experimentation and observation. By engaging in this hands-on learning process, you'll gain confidence in creating accurate and visually appealing isometric drawings that effectively represent three-dimensional objects on a two-dimensional surface.

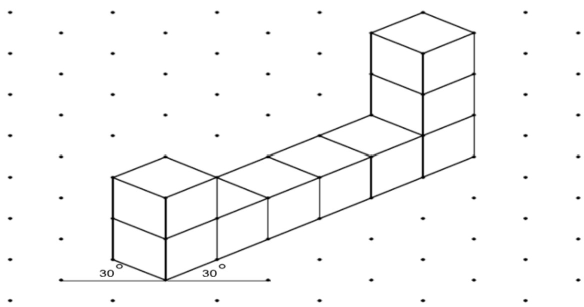

Figure 3: Example Isometric Drawing

4.2 Orthographic Drawing

Orthographic drawings typically consist of three views: top, front, and right side. The front view is chosen based on the side that provides the most comprehensive information about the object. When an edge lies inside the object and is not visible from the outside, it is represented by a line in the drawing. These technical drawings are essential in engineering, architecture, and design, allowing precise visualization and understanding of an object's shape and dimensions from different viewpoints. By presenting multiple views of an object, orthographic drawings provide comprehensive and accurate information for manufacturing, construction, and other applications, ensuring precise execution of projects and minimizing errors.

4.2.1 Construction

When creating an orthographic drawing of an object, a useful technique is to envision the object unfolding like a cube before starting the drawing. This approach aids in accurately depicting the object's dimensions and relationships in the orthographic view. For instance, in the given example, the front view typically provides essential details that are more critical than those in the top and side views. To ensure coherence between the views, the edges in the top and side views align with the corresponding edges in the front view. This alignment is essential for maintaining accuracy and consistency in the orthographic drawing. By following these practices, engineers and designers can produce clear and comprehensive orthographic drawings that serve as valuable references for construction, manufacturing, and various design processes.

Mastering orthographic drawing is best achieved through hands-on experience similar to isometric drawings. Like isometric drawings, start by creating orthographic drawings. Take a few of your isometric drawings and assess which face provides the most information about the object; this will be your front face. Once the front face is identified, proceed to deconstruct the object into three views: top, front, and right side. Ensure that these different views align with the front view, as demonstrated in the example. This alignment is crucial to maintaining accuracy and consistency in the orthographic drawing, allowing for a comprehensive representation of the object's dimensions and features from multiple angles. By actively engaging in this practice, you'll enhance your proficiency in orthographic drawing, a valuable skill in various engineering and design disciplines.

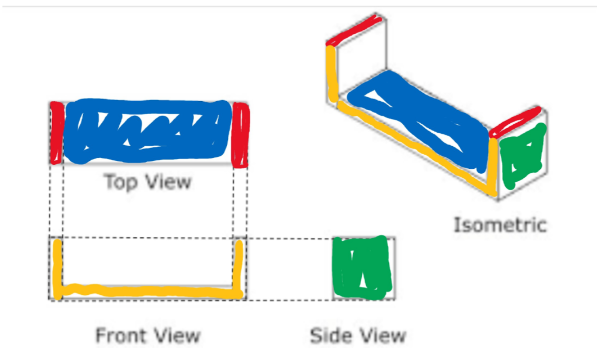

Figure 4: Example Orthographic Drawing

Drawing Conclusion

In conclusion, this lesson on isometric and orthographic drawings provides participants with a valuable opportunity to collaborate closely with the CAD subdivision and gain practical experience in interpreting technical drawings. By distinguishing between isometric and orthographic schematic styles, participants learn to accurately interpret and translate these drawings into tangible components for engineering and design projects.

Isometric drawings offer a three-dimensional representation of objects without distortion, allowing for realistic and easily understandable visualisations. This technique is essential in engineering, architecture, and design for effective communication and problem-solving. On the other hand, orthographic drawings present multiple views of an object, including top, front, and right side, providing comprehensive and precise information for manufacturing and construction processes.

Throughout the lesson, participants develop proficiency in interpreting intricate details conveyed in these technical drawings, enabling them to recreate components with precision and efficiency. This skill is invaluable for engineering and design professionals as it ensures accurate implementation of designs and optimises the manufacturing or construction process.

By embracing hands-on practice in both isometric and orthographic drawing, participants can deepen their understanding and mastery of these essential skills. Engaging in practical exercises and experimenting with various objects enhances their ability to create accurate and visually appealing drawings, contributing to successful projects and effective communication in the fields of engineering and design. Overall, the lesson equips participants with valuable tools and expertise to excel in their future endeavours and contribute meaningfully to various industries.

5 Strategic Design

Strategic design plays a vital role in constructing a successful and efficient robot. This stage involves meticulous planning, where teams determine the most effective subsystems to incorporate into their robot and devise optimal strategies for building them. Participants in this process will gain valuable skills in analysing the game manual and understanding the intricacies of the competition. By thoroughly examining the rules and objectives, teams can strategize and align their robot design with the game's specific challenges and scoring opportunities.

Analysing the game is a critical aspect of strategic design. Teams will assess the game's dynamics, scoring mechanisms, and potential obstacles to identify key priorities and opportunities for maximising points. By understanding the game's nuances and strategies, teams can make informed decisions about which subsystems to implement and how to optimise their performance to excel in various game scenarios.

The process of strategic design empowers teams to make well-informed choices in designing their robot, setting them up for success in the competition. The ability to strategize and plan efficiently not only enhances the robot's overall effectiveness but also provides a competitive edge in the competition. Through this stage, participants gain invaluable experience in utilising resources wisely and aligning their robot's capabilities with the game's requirements, ultimately positioning themselves for a strong performance and potential victory.

5.1 Design for FRC

Designing in FRC adheres to "golden rules" essential for achieving success in creating subsystems and robots. These rules serve as guiding principles, ensuring that the design process is effective and efficient. Participants in FRC will learn and apply these rules throughout the design stages to optimize their robot's performance and chances of success in the competition.

The "golden rules" encompass various aspects of robot design, including functionality, durability, weight management, and adherence to the game's rules and constraints. Engineers and teams aim to design subsystems that not only accomplish the required tasks but also function seamlessly and reliably during the intense competition.

Weight management is a critical consideration in FRC design. Teams must balance the robot's capabilities with weight limitations to ensure compliance with competition rules and maximize performance. By carefully evaluating the weight distribution and minimizing unnecessary mass, teams can optimize their robot's agility and maneuverability.

Following these "golden rules" fosters a systematic and disciplined approach to robot design. By incorporating them into the design process, teams can increase their chances of creating a highly competitive robot, equipped to excel in the dynamic and challenging environment of the FRC competition.

5.2 Golden Rules

The "golden rules" in the world of FRC (FIRST Robotics Competition) are a set of essential guidelines crafted from the collective experiences and knowledge of prominent teams, specifically FRC teams 1114, 2056, and 1678. These teams have generated 5 "golden rules" that were derived from different sources: Design within your limit, design for quality, keep design simple. Reuse proven designs, and design with the drivers in mind

5.2.1 Design Within Your Limits

When embarking on the robot design process, it is crucial to consider and align with your team's specific capabilities and resources. Three primary factors to keep in mind are your team's manufacturing, programming, and CAD (Computer-Aided Design) abilities. Understanding these strengths and limitations enables you to tailor your design to what your team can effectively produce and implement.

Manufacturing capabilities encompass the skills and tools available to construct physical components. By recognizing your team's expertise in manufacturing, you can design subsystems and mechanisms that align with your team's proficiency, ensuring a smoother and more efficient build process.

Programming abilities are equally important in robot design. Consider the level of expertise within your team when designing features that require complex coding and autonomous functionalities. By designing within your team's programming capabilities, you can ensure that your robot's software aligns with your team's potential to implement and optimize its performance.

CAD abilities play a significant role in visualizing and planning your robot's design. Leveraging CAD software effectively allows you to create detailed and accurate representations of your robot before manufacturing. By understanding your team's CAD capabilities, you can leverage this valuable tool to optimize your design and improve overall efficiency.

Furthermore, it is essential to keep your team's budget in mind during the design process. Design choices should align with the financial resources available to your team. Prioritize cost-effective solutions and consider the availability of Commercial Off-The-Shelf (COTS) parts that fit within your budget constraints.

By considering these key factors throughout the design process, you can create a well-informed and practical robot design that maximize your team's strengths while ensuring cost-effectiveness and feasibility within your resources. Emphasizing adaptability and leveraging your team's capabilities will lead to a more successful and competitive robot in the FRC competition.

5.2.2 Design for Quality Over Quantity

Designing for quality over quantity is a fundamental principle that prioritizes excellence in a robot's performance. This concept emphasizes the importance of creating subsystems and mechanisms that deliver consistent and reliable results, even if it means sacrificing quantity. For instance, consider a shooter mechanism: it is preferable to design one that is 95% accurate and shoots a smaller number of balls (3-4) at a time, rather than a shooter that is 64% accurate but can shoot more balls (7-8) at once. By focusing on accuracy and reliability, teams can increase their robot's overall effectiveness and scoring potential during the competition.

Reliability is a key factor that can significantly impact a team's alliance selection in the FRC competition. While the top six teams often have an advantage in alliance selection, teams that demonstrate high reliability and consistency in their robot's performance stand a strong chance of being chosen by higher-ranked teams as alliance partners. Being a reliable and dependable partner during matches can make a team an attractive choice for alliance captains looking for consistency and teamwork to maximize their chances of success in the playoffs.

By embracing the principle of designing for quality over quantity, teams prioritize delivering top-notch performance in specific tasks, ultimately contributing to better overall match outcomes and strategic alliance selections. Quality design choices that emphasize accuracy and reliability play a pivotal role in positioning teams for competitive success and fostering strong collaboration with other high-performing teams during the FRC competition.

5.2.3 Keep Designs Simple

Simplicity in subsystem designs is a key principle that brings numerous benefits to the robot design process. By keeping designs straightforward, teams can reduce the risk of complications and malfunctions during the competition. The "less moving parts" rule of thumb guides this approach, as it decreases the chances of mechanical issues and failures, resulting in a more robust and reliable robot.

In addition to enhancing reliability, simplicity fosters better comprehension of the subsystem's functionality and operation. Teams and drivers can more easily understand how the subsystem interacts with the rest of the robot, facilitating effective strategize and execution during matches.

Practically, simple designs are quicker and easier to manufacture, optimizing teams' time and resources. Furthermore, in case of a subsystem malfunction, troubleshooting and repairs become more manageable and efficient. This leads to quicker turnaround times during pit stops, maximizing the robot's up time and overall performance in the dynamic and competitive FRC environment. Embracing simplicity in subsystem designs empowers teams to build more efficient, reliable, and competitive robots.

5.2.4 Reuse Proven Designs

The analogy "Don't reinvent the wheel" is frequently employed to highlight the importance of leveraging existing solutions rather than starting from scratch. In the FRC context, each year's kit of parts includes a chassis that serves as a well-designed drive base. Customizing and building upon proven designs is often a more efficient and productive use of time compared to creating entirely new designs.

Utilizing the pre-existing chassis reduces the risk of encountering issues with untested designs, ensuring that the robot's foundational structure is reliable and functional. It saves valuable time during the limited build season, allowing teams to focus on refining and enhancing specific subsystems and functionalities that align with the year's game objectives.

By avoiding unnecessary reinvention, teams can maximize their productivity, allocate resources effectively, and streamline the build process. This approach not only minimizes potential design flaws and setbacks but also optimizes the team's chances of developing a competitive and well-performing robot within the tight time frame of the FRC competition. Embracing proven designs sets teams on a path to success, fostering innovation in targeted areas while maintaining a reliable and effective foundation for their robot.

5.2.5 Design with Drivers in Mind

Designing with the drivers in mind is a critical consideration in FRC robot design. The drivers are the ones who will operate the robot during competition, making it essential to create a user-friendly and intuitive control system that enhances their performance. To optimize the drivers' scoring cycle time, it is crucial to make tasks easier for them to accomplish.Key tasks that directly impact scoring cycle time include:

1. **Navigating to pickup areas**: Designing the robot's mobility and control to make it easy for drivers to maneuver and reach different pick-up locations on the field.

2. **Lining up with game pieces for pickup**: Incorporating features that aid drivers in accurately aligning the robot with game pieces, ensuring efficient and swift pick-up actions.

3. **Grabbing game pieces**: Creating mechanisms that enable drivers to smoothly and reliably grasp game pieces without unnecessary complications or delays.

4. **Having control while moving**: Designing subsystems that allow drivers to maintain precise control over game pieces while the robot is in motion, preventing mishaps or fumbles during game play.

5. **Piece Dropoff**: Designing the robot's drivetrain and manipulators to facilitate easy positioning for smooth and accurate game piece delivery.

By prioritizing the drivers' ease of operation and providing them with the necessary tools and capabilities, teams can optimize their scoring cycle time, enhancing their robot's overall performance in the competition. A well-designed and driver-centric robot can significantly impact a team's success by increasing scoring efficiency and contributing to strategic gameplay.

Strategic Design Conclusion

In conclusion, the "golden rules" serve as essential guidelines for successful robot design in the world of FRC. Designing within your team's capabilities, considering manufacturing, programming, and CAD abilities, ensures a practical and feasible robot design. By embracing simplicity in subsystem designs and keeping moving parts to a minimum, teams can enhance reliability and ease of troubleshooting, leading to more efficient and competitive robots.

Furthermore, prioritizing quality over quantity in robot design fosters consistent and reliable performance, making teams attractive alliance partners during alliance selections. Leveraging existing proven designs, such as the kit of parts chassis, allows teams to streamline the build process, save time, and focus on refining specific subsystems to meet the game's objectives effectively.

Finally, designing with the drivers in mind plays a pivotal role in enhancing the robot's overall performance. By creating user-friendly and intuitive control systems, teams can optimize scoring cycle time and empower their drivers to operate the robot with precision and efficiency during the FRC competition. Embracing these golden rules and applying them throughout the design process sets teams on a path to success, empowering them to create well-performing and competitive robots that excel in the dynamic and challenging FRC environment.

6 Usefulness

To increase the value of your alliance, it is crucial to maximize overall points during each phase of the game: autonomous, tele-op, and endgame. Coordinating strategic actions across all periods can lead to greater scoring opportunities and a more competitive alliance.

An illustrative example is collecting game pieces at the end of the autonomous period. The game pieces gathered during this time would carry over to the start of the tele-op period, providing an advantageous head start. By efficiently collecting and storing game pieces in the autonomous period, the robot gains a valuable advantage in positioning and scoring opportunities when the tele-op period commences.

Taking a holistic approach to scoring ensures that the team can excel in all game phases, complementing the efforts of alliance partners and maximizing the alliance's collective performance. Effectively coordinating actions between periods and strategize based on game dynamics enables the alliance to adapt and optimize its scoring potential throughout the match.

By maximizing scoring potential across all phases of the game, teams can become invaluable alliance partners, contributing significantly to the alliance's overall success. Strategic coordination and efficient use of game resources lead to a higher cumulative score, positioning the alliance for competitive advantage and success in the FRC competition.

Usefulness Conclusion

To form a strong alliance, teams must maximize scoring opportunities in all game phases: autonomous, tele-op, and endgame. Efficiently collecting and storing game pieces during autonomous can give a valuable head start for scoring in tele-op. By coordinating actions and adapting to game dynamics, alliances can boost their cumulative score and increase their chances of success in the FRC competition.

7 Cost Benefit Analysis

When conducting a cost-benefit analysis, there are several key points to consider. Evaluating the benefits of one design over another and the resources required to implement a subsystem is essential. Several factors can be used to judge the worthiness of a subsystem:

1. **Time to line up with the target**: Assessing how efficiently the robot can align itself with the target for scoring actions can impact overall match performance and point accumulation.

2. **Calibration difficulty**: Understanding the complexity of calibrating shooting mechanisms versus placement-based subsystems can influence the overall accuracy and consistency of scoring actions.

3. **Allowable deviation before missing**: Determining the maximum acceptable error margin the robot can have before it misses its intended target is crucial for optimising scoring opportunities.

4. **Engineering difficulty**: Evaluating the complexity and technical challenges involved in designing and building the subsystem provides insight into the overall feasibility and implementation effort required.

5. **Prototyping difficulty**: Considering the ease of prototyping the subsystem helps in assessing

how quickly and effectively the team can iterate and refine the design during the development phase.

6. **Coding requirements**: Understanding the amount of programming and control logic needed to operate a more complicated subsystem with many moving parts can impact development time and resources.

7. **Existing subsystem availability**: Examining whether the desired subsystem already exists as a proven design or can be adapted from previous successful implementations, as discussed in Section 4.2.4 - Reuse Proven Design, may save time and resources.

By thoroughly analysing these factors during the cost-benefit analysis, teams can make informed decisions about the most viable subsystem design options. This systematic approach helps ensure that the team's efforts are well-directed, resources are optimised, and the robot's overall capabilities are optimised for competitive success in the FRC competition.

Cost Benefit Analysis Conclusion

Conducting a comprehensive cost-benefit analysis is crucial for making informed decisions in robot design. Evaluating factors like alignment efficiency, calibration difficulty, allowable deviation, and engineering complexity helps optimise subsystem choices. By systematically considering these elements, teams can ensure resource optimization and enhance the robot's competitive success in the FRC competition.

8 Math

Math is, to no one's surprise, very important. Here we are going to talk about the math behind many principles that go into designing a robot such as gear ratios, torque, mechanical advantage, and more however it is important to note that even though we learn many ways of hand calculating the answer there is often spreadsheets that work to automate these mundane calculations. Learning the principles behind this will aid you when designing subsystems and mechanisms.

8.1 Gear Ratios

Gear ratios play a crucial role in the design of various mechanisms in the robot. A gear ratio is established when two gears are connected to each other: the gear driven by a motor or similar component is referred to as the "driver gear," while the gear that rotates due to the motion of the driver gear is called the "driven gear."

To determine the gear ratio, divide the number of teeth on the driven gear by the number of teeth on the driving gear. This ratio provides valuable information: each time the driven gear completes a full rotation, the driving gear rotates by the calculated amount based on the gear ratio.

8.1.1 Example

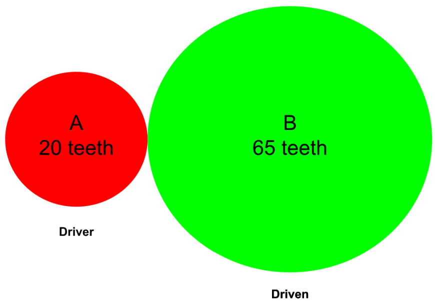

Figure 5: Example Gear Ratio

$$\frac{Driven}{Driver} = \frac{65}{20} = 3.25$$

Interpretation: Every time Gear B makes a complete rotation, Gear A makes 3.25 complete rotations

Calculating and understanding gear ratios is essential for achieving specific functionalities in the robot's mechanical systems. By choosing appropriate gear sizes and ratios, teams can optimise speed, torque, and mechanical advantage in various subsystems, ensuring efficient and precise movements during the FRC competition.

Gear ratios are a fundamental aspect of mechanical design, and their proper utilisation empowers teams to engineer robots that meet the specific challenges of the game and excel in their intended tasks. This mathematical approach to gear ratios leads to better-designed robots, offering higher performance and increased potential for success on the field.

Gear Ratio Conclusion

In conclusion, understanding and calculating gear ratios are essential skills in the design of effective mechanical systems for FRC robots. Gear ratios are determined by the number of teeth on the driven and driver gears, and they play a crucial role in optimising speed, torque, and mechanical advantage in various robot subsystems.

By carefully choosing appropriate gear sizes and ratios, teams can tailor their robots' mechanical systems to meet the specific challenges of the game and excel in their intended tasks during the FRC competition. Gear ratios enable efficient and precise movements, contributing to the overall performance and success of the robot on the field.

The mathematical approach to gear ratios provides a solid foundation for designing well-optimised robots with higher performance capabilities. By mastering gear ratio calculations, teams can fine-tune their mechanical designs, ensuring that their robots are equipped with the right balance of speed and power to perform at their best in the FRC arena. As teams continue to extend their thinking and explore various gear combinations, they gain valuable insights into how different gear ratios impact the robot's functionality, enabling continuous improvement and innovation in their designs.

8.2 Angles and Parabolas

Designing shooters requires careful consideration of several factors to ensure accurate and effective performance. An essential aspect is determining the impact point of the shooter and exploring ways to correct the shot if it misses the target. This process involves understanding the shooter's trajectory and identifying potential adjustments to enhance shooting accuracy.

The angle at which the shooter is positioned depends on multiple factors, such as the shooter's power, the robot's position relative to the target, and the target's location. If the robot is closer to the target, the shooter may need to be angled higher, while a more distant target might require a lower angle. Calculating the optimal shooting angle based on these variables is crucial for achieving precise and consistent results.

When the shooter fires, the projectile follows a downwards parabolic path due to gravity's effect. The trajectory of the parabola varies depending on the shooter's angle and power. By understanding and analysing this parabolic trajectory, teams can make informed decisions about shooter adjustments and optimise the robot's shooting capabilities.

Considering these aspects during the shooter's design phase allows teams to fine-tune the shooter's performance, improving accuracy and efficiency. Implementing corrective measures based on the trajectory analysis enhances the robot's scoring potential and contributes to a more competitive and successful performance in the FRC competition.

8.2.1 Flywheel Shooter

8.2.1.1 Single Wheel

Uses one spinning wheel to launch game pieces.

8.2.1.2 Double Wheel

Uses two wheels spinning in opposite directions to grip and launch game pieces, providing better control and consistency.

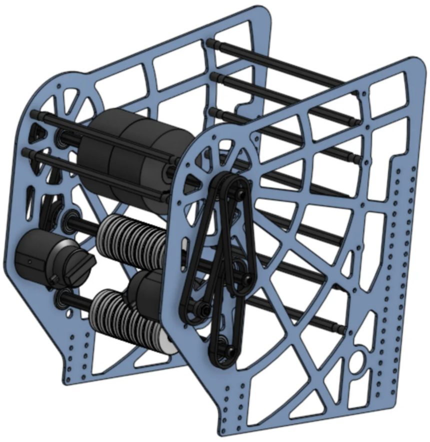

Figure 6: Single Wheel Shooter
Source: [**falcon2022cd**]

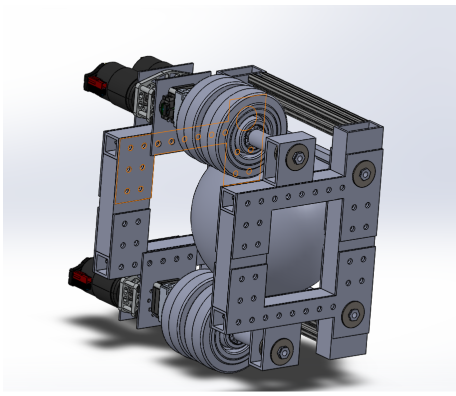

Figure 7: Double Wheel Shooter
Source: [**pickleb2020cd**]

8.2.2 Catapult Shooter

Utilizes a mechanical arm or lever to launch game pieces by releasing stored energy (e.g., springs, surgical tubing).

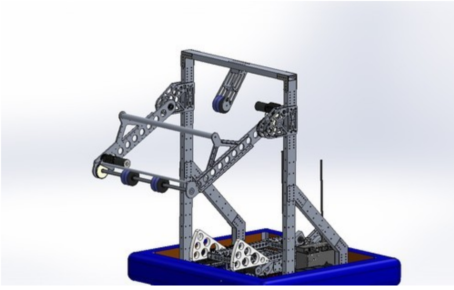

Figure 8: Catapult Shooter
Source: [**scheiber2014**]

8.2.3 Puncher Shooter

Uses a linear actuator or pneumatic piston to punch or push game pieces outwards quickly.

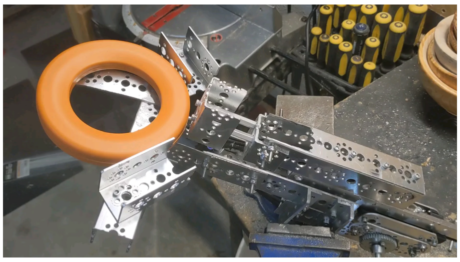

Figure 9: Puncher Shooter
Source: [**jimmy12091**]

8.2.4 Turrent Shooter

A shooter that can rotate (usually 360 degrees) to aim at targets more accurately without needing to move the entire robot.

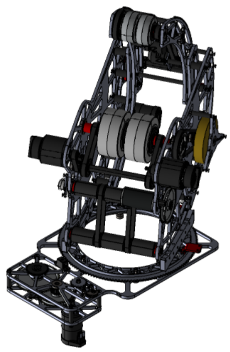

Figure 10: Turrent Shooter
Source: [**jimmy12091**]

9 Measuring Tools

In engineering and design, the accurate measurement of objects is essential, and various tools are available for this purpose. Each tool possesses unique characteristics that make it suitable for specific measurement tasks. For instance, a digital calliper is known for its precision, allowing for highly accurate measurements, while a ruler may offer greater versatility in measuring different types of objects.

This lesson aims to familiarise learners with the usage of devices like the digital calliper to achieve precise measurements. Mastering the digital calliper and similar tools empowers individuals to obtain accurate dimensions critical for engineering projects and ensures precise fits and alignments in the construction of robot components.

By understanding how to use precise measurement tools effectively, students can enhance their engineering skills, ensure adherence to design specifications, and produce high-quality components that contribute to the success of their FRC robots. This emphasis on precise measurement techniques is a fundamental aspect of successful engineering design, enabling teams to optimise the performance and reliability of their robots during the competition.

9.1 Calliper

A digital calliper is a valuable electronic measuring device known for its relatively high level of accuracy, typically providing measurements with a margin of error around ±0.02 mm or ±0.001 in. However, it is important to note that the margin of error can vary based on the length of the calliper being used.

This versatile tool is widely used in engineering and manufacturing industries, offering precision in measuring various dimensions. Its digital display provides quick and easy-to-read measurements, enabling efficient and reliable data collection for designing and constructing robot components.

Understanding the capabilities and limitations of a digital calliper is essential for obtaining accurate measurements in engineering projects. By being aware of the potential variations in the margin of error, users can apply appropriate techniques and best practices to achieve the desired level of precision in their measurements.

Incorporating the digital calliper as a primary measuring tool in the design and construction processes empowers teams to ensure the dimensional accuracy and fit of robot components, contributing to the overall performance and success of their FRC robots in the competition.

9.2 Measuring Tape

A measuring tape is a versatile and commonly used tool for taking linear measurements. It consists of a flexible strip, usually made of metal or cloth, marked with standardised measurements such as inches, centimetres, or both. Measuring tapes come in various lengths, making them suitable for a wide range of measurement tasks.

This tool is widely utilised in various industries, including construction, engineering, and design. Its flexibility allows for easy measurement of both straight and curved surfaces, making it an ideal choice for projects that involve irregular shapes or contours.

In the context of FRC, a measuring tape can be invaluable for tasks such as determining the dimensions of game elements, ensuring precise alignments during robot assembly, and assessing spatial requirements for various subsystems.

By including a measuring tape in the toolkit, teams can accurately measure lengths and dimensions, aiding in the creation of well-designed and precisely fabricated robot components. The versatility and convenience of a measuring tape make it an indispensable tool for any FRC team, contributing to the successful development and construction of their robots for the competition.

9.3 Measuring Tools Conclusion

In conclusion, accurate measurement is a critical aspect of successful engineering and design, particularly in the construction of FRC robots. The use of precise measurement tools, such as digital callipers and measuring tapes, empowers learners to obtain accurate dimensions and ensure precise fits and alignments in their robot components. Understanding the capabilities and limitations of these tools enables teams to achieve the desired level of precision in their measurements, contributing to the overall performance and success of their robots during the competition.

Mastering the digital calliper and incorporating it as a primary measuring tool in the design and construction processes allows for high-level accuracy, ensuring adherence to design specifications. On the other hand, the versatility and convenience of a measuring tape make it an indispensable tool for measuring both straight and curved surfaces, aiding in various tasks, such as determining game element dimensions and assessing spatial requirements for subsystems.

By emphasising the importance of accurate measurement techniques and utilising precise measuring tools effectively, FRC teams can enhance their engineering skills and produce high-quality components. These practices ultimately contribute to the success of their robots and help them perform optimally during the competition.

10 Safety

Safety is a paramount aspect of the FRC experience, emphasising the importance of safe practices in manufacturing and within the FIRST community. Whether it's attending events, building the robot, or hosting activities, safety is a top priority at all times.

In this critical lesson, participants will learn essential rules and regulations to ensure safety while handling power tools, equipment, and during various activities. Emphasis will be placed on the proper use of personal protective equipment such as face and eye protection to minimise potential risks and injuries.

Furthermore, the lesson will underscore the significance of safety protocols specific to the robot's construction. Understanding and implementing safety measures when working with the robot and its components are vital to prevent accidents and foster a secure environment for all team members. Adhering to safety guidelines not only safeguards the well-being of team members but also ensures a positive and secure experience in the FRC community and at events.

10.1 Face and Eye Protection

Safety is of utmost importance when it comes to face and eye protection in various settings. A wide array of protective equipment is available, ranging from safety glasses to more specialised welding shields. Taking necessary precautions to safeguard the eyes and face is essential to prevent potential hazards.

In this context, a crucial practice is to inspect the protective equipment for any signs of damage or defects after each use. Regularly checking the equipment ensures that it remains in optimal condition and can effectively protect against potential hazards in subsequent tasks or activities.

By diligently inspecting face and eye protection equipment, users can identify any issues or wear and tear that might compromise its effectiveness. This proactive approach to safety contributes to maintaining a secure working environment and reduces the risk of accidents or injuries during manufacturing, construction, or any other relevant activities.

Ultimately, prioritising the use of appropriate face and eye protection, coupled with routine inspection and maintenance, enhances the safety and well-being of individuals in diverse settings, including FRC activities and beyond. Emphasising safety practices promotes a culture of responsibility and ensures a positive experience for all team members while mitigating potential risks in various environments.

10.1.1 Use and Application

Safety glasses are essential protective equipment when engaging in various tasks, especially those involving work on the robot, such as grinding, drilling, soldering, cutting, welding, and other related activities. These glasses provide crucial eye protection against potential hazards like flying objects or exposure to chemicals, such as splatter, splashes, or sprays.

At FIRST events, wearing safety glasses is equally vital to maintain a safe environment. In the pits, near the arena or playing field, and even on the practice field, safety glasses should be worn as a precautionary measure. Anytime signs indicate the need for safety glasses, it is crucial to adhere to the safety protocols and wear them promptly.

By consistently wearing safety glasses during robot work and at FIRST events, participants can effectively shield their eyes from potential dangers and reduce the risk of eye injuries. Safety glasses play a significant role in fostering a safe and responsible environment, instilling good safety habits among team members, and setting a positive example for others in the FIRST community.

Promoting a culture of safety with the use of safety glasses not only safeguards participants' well-being but also reinforces the importance of safety awareness and practices throughout the FRC program and beyond. Emphasising eye protection serves as a proactive step toward preventing accidents and maintaining a secure and enjoyable experience for everyone involved.

Use safety glasses when:

- You are performing any work on the robot from grinding, drilling, soldering, cutting, welding, e.t.c.
- There is a risk of flying objects or exposure to chemicals such as splatter, splashes, and/or sprays

Wear safety glasses at FIRST events when:

- You're anywhere in the pits
- You're anywhere near the arena including the playing field
- You're on the practice field
- You see signs indicating you need safety glasses

10.1.2 Safety Glasses and Protection Eyewear

Safety glasses and protective eyewear play a critical role in creating a protective barrier around the eyes, shielding them from potential hazards like flying projectiles, splashes, and compressed air, among other risks. To ensure injury prevention, it is mandatory for all individuals present in the pit, practice arena, and playing field to wear safety glasses or protective eyewear that meet specific safety standards.

These safety standards include ANSI approval, UL listing, CE EN166 rating, AS/NZS certification, or CSA rating. Compliance with these rigorous safety standards ensures that the glasses provide adequate protection to the wearers.

For FIRST-approved safety glasses, only lightly tinted yellow, rose, blue, or amber tints are permissible. These tints are carefully selected to ensure visibility while maintaining safety. Additionally, it is essential that the wearers' eyes remain visible to others, enhancing communication and fostering a safer environment.

The restriction on using any other safety glasses or protective eyewear highlights the significance of adhering to approved standards. By strictly enforcing the use of certified safety glasses, FIRST emphasises the commitment to safety and ensures that all participants can confidently engage in activities while minimising potential risks.

Promoting the use of properly rated and FIRST-approved safety glasses underscores the organisation's dedication to safety and fosters a culture of responsibility within the FRC community. Prioritising eye protection not only reduces the risk of injuries but also reinforces the values of safety awareness and compliance in all aspects of the program.

10.1.3 Face and Eye Safety Conclusion

In conclusion, safety is paramount when it comes to face and eye protection in various settings, including FRC activities. Regularly inspecting protective equipment for damage and defects enhances its effectiveness in safeguarding against potential hazards. Emphasising the use of appropriate face and eye protection, especially safety glasses, during tasks and at FIRST events contributes to a secure working environment and reduces the risk of accidents. Complying with safety standards for approved safety glasses reinforces a culture of responsibility and safety consciousness within the FRC community. By prioritising eye protection and adhering to safety protocols, individuals can confidently participate in various activities while minimising risks and ensuring a positive and safe experience for all team members.

10.2 Robot Safety

Understanding how to ensure safety when interacting with the robot is of utmost importance to prevent injuries and accidents. Whether transporting the robot, working on it, or simply being around it, implementing proper safety measures is essential.

Being aware of potential hazards and practising safety protocols while handling the robot significantly reduces the risk of injuries. This includes understanding the robot's movement and ensuring clear paths during transportation, as well as following safety guidelines during any maintenance or adjustments.

Maintaining a safe distance and wearing appropriate protective gear when working on the robot further contributes to injury prevention. By fostering a safety-conscious mindset, individuals can create a secure environment for themselves and others when in close proximity to the robot.

Emphasising safety awareness when interacting with the robot is a fundamental aspect of responsible robot handling in the FRC community. This knowledge not only safeguards team members but also instils a culture of safety that extends to all robot-related activities. Prioritising safety while working around the robot enhances the overall experience and ensures that participants can fully engage in robot-related tasks with confidence and peace of mind.

10.2.1 Transportation

Knowing how to move the robot properly and safely is crucial to prevent potential injuries in the future. Practising safe techniques when moving the robot significantly reduces the risk of accidents resulting from mishandling. By implementing safe practices during robot transportation, the likelihood of injuries occurring decreases, enhancing the overall safety of the team.

For longer distances, it is strongly advised to use a robot cart or have multiple team members push the robot together. This helps distribute the weight and ensures a safer transfer.

Before lifting the robot, a comprehensive checklist must be completed to ensure a safe lifting process. This includes ensuring that everyone involved has the proper personal protective equipment (PPE) and verifying essential aspects like whether the robot is powered, all parts are secured, and no one is currently working on the robot. Ideally, 2 to 4 people should be involved in lifting the robot to minimise strain.

Additionally, clear communication among team members is crucial. Everyone should be aware of the destination and the path they will take when moving the robot. It is also essential to ensure that the path and surrounding area are free of obstacles and debris, reducing the risk of tripping or stumbling during the transfer.

By consistently adhering to safe moving practices, teams can create a culture of safety and responsibility. These precautions help protect team members from potential injuries, ensuring that the focus remains on successfully building and competing with the robot during the FRC season. See below for steps to lift a robot.

Checklist to complete before lifting the robot:

1. Make sure everyone has proper PPE
2. Before moving double check for:
    (a) Is the robot powered?
    (b) Are the parts of the robot secured?
    (c) Is anyone working on the robot?
    (d) Do you have enough people to lift the robot(2 - 4 people preferred)
3. Make sure that everyone knows where they are taking the robot and the path they are going to take

4. Make sure that the path and area you are taking the robot is free of obstacles and debris

10.2.2 Robot Cart

Robot carts play a crucial role in safely transporting robots, and there are specific rules that govern their usage. To comply with regulations and maintain a secure environment, the following rules apply to robot carts:

1. Robot carts must be present in the pits when they are not actively transporting a robot. This ensures that the robot is securely stored when not in use and minimises the risk of accidents or damage.

2. Robot carts must have dimensions that allow them to fit through a standard 30-inch door. This ensures ease of mobility and accessibility during transportation.

3. Robot carts must be designed in a way that does not cause damage to the site's flooring. This precaution helps maintain the venue's infrastructure and ensures a safe environment for all participants.

4. Music or sound-generating devices on the robot carts are not allowed, except for devices that emit sounds at a reasonable volume to alert others nearby that a robot is in motion. This promotes a noise-conscious environment while ensuring everyone's awareness of robot movement.

5. Each robot cart must prominently display the team number, making it easily identifiable by field personnel and other team members. This aids in the organisation and management of robot carts during competitions and events.

By adhering to these rules, teams can ensure the safe and efficient transportation of robots during the FRC season. Emphasising compliance with cart regulations fosters a culture of responsibility and safety, enhancing the overall experience for all participants while protecting the robot, the venue, and the well-being of everyone involved.

10.2.3 Working on the Robot

When engaging in work on the robot, prioritising safety is essential, and this involves ensuring that all necessary safety equipment, proper personal protective equipment (PPE), and safety precautions are in place. One of the fundamental safety measures is the mandatory use of safety glasses, regardless of whether team members already wear prescription glasses. These glasses provide crucial eye protection during various tasks and help minimise the risk of potential eye injuries.

In addition to safety glasses, safety gloves may also be required in certain situations when working on the robot. These gloves offer protection to team members' hands during tasks that involve sharp edges, rough surfaces, or potential contact with hazardous materials.

By diligently adhering to safety guidelines and wearing appropriate safety gear, team members can significantly reduce the risk of injuries and accidents while working on the robot. These safety practices promote a culture of responsibility and ensure a safe and secure environment for everyone involved in the FRC program. Emphasising safety precautions and providing proper PPE demonstrates the team's commitment to the well-being of its members, enhancing the overall experience and productivity during robot assembly and maintenance.

Before working on the robot, double check to make sure:

- Loose drawstrings tucked away
- Long hair tied
- You have necessary PPE equipment

- Foot protection is worn(No open toed footwear permitted when working on or around the robot)

10.2.4 Robot Safety Conclusion

In conclusion, safety is paramount when interacting with the robot, and it is crucial to follow proper safety measures to prevent injuries and accidents. Understanding potential hazards and implementing safety protocols during robot transportation, maintenance, and assembly significantly reduces the risk of harm. Prioritising safety through clear communication, wearing appropriate protective gear, and adhering to safety guidelines fosters a culture of responsibility and ensures a secure environment for all team members. By consistently emphasising safety practices, teams can confidently engage in robot-related tasks while protecting their well-being and promoting a safe and productive FRC experience.

10.3 Power Tool Safety

Safety around power tools is a top priority and receives significant attention and training. If you have any questions or concerns about operating a particular tool or suspect that the equipment may be unsafe (e.g., damaged, broken, or missing pieces), it is crucial to seek assistance from someone else experienced with the tool. Alternatively, you can contact your manufacturing lead if you believe that a tool is no longer safe to operate. Prioritising safety in this manner ensures that team members are well-informed and can confidently utilise the tools without compromising their well-being.

As always, safety glasses are a non-negotiable requirement. You will consistently find the need for safety glasses when building, assembling, and moving the robot or any workpiece. The consistent use of safety glasses protects team members' eyes from potential hazards and is an essential safety measure throughout various activities.

The use of safety gloves, on the other hand, depends heavily on the specific type of tool being used. When using certain power tools that involve sharp edges or hazardous materials, safety gloves may be necessary to protect team members' hands during the tasks. Careful consideration of the type of tool and the associated risks will determine whether safety gloves are required.

By emphasising the importance of safety gear and adhering to safety guidelines, team members can confidently work with power tools while minimising the risk of injuries. These safety practices promote a culture of safety consciousness and responsibility, ensuring a safe and productive environment for everyone involved in the robot's construction and assembly.

10.3.1 Safety Gloves

Certain tools, such as the deburring tool, soldering iron, hand saw, and angle grinder, require the use of safety gloves to ensure the safety of team members during their operation.

1. **Deburring tool**: When using a deburring tool to remove sharp edges or burrs from metal or plastic, safety gloves provide an extra layer of protection for hands, guarding against potential cuts or abrasions.

2. **Soldering iron**: Soldering involves working with high temperatures and molten metal. Safety gloves shield hands from accidental contact with the hot surfaces and the risk of burns.

3. **Hand saw**: Hand saws, especially those with sharp teeth, can pose a hazard to hands during cutting tasks. Safety gloves offer added protection against cuts or injuries while using these tools.

4. **Angle grinder**: Angle grinders are powerful tools that require caution during use. Safety gloves provide protection from sparks, abrasive materials, and potential cuts or abrasions.

By utilising safety gloves while operating these tools, team members minimise the risk of hand injuries and create a safer work environment. Emphasising the importance of safety gear and adhering to safety protocols during the use of these tools demonstrates a commitment to team members' well-being and enhances the overall safety culture within the FRC community.

10.3.2 No Safety Gloves

Certain tools, such as the drill press and other rotating equipment, do not require the use of safety gloves. In fact, wearing gloves while operating these tools may increase the risk of serious injuries, such as getting pulled into the machine, potentially leading to amputation or even death.

1. **Drill press**: The drill press is a stationary tool used for precise drilling operations. When operating a drill press, it is essential to avoid wearing gloves, as they can become entangled in the rotating parts, posing a significant safety hazard.

2. **Rotating equipment**: This includes various machinery with rotating parts, such as lathes, milling machines, and grinders. When working with any rotating equipment, wearing gloves is discouraged to prevent any entanglement incidents and ensure safe operation.

The decision to avoid wearing gloves while operating rotating equipment is rooted in the principle of safety. Gloves can be caught in moving parts, leading to severe injuries. By refraining from wearing gloves when using these tools, team members reduce the risk of accidents and create a safer work environment.

10.3.3 Power Tool Safety Conclusion

In conclusion, safety is paramount when working with power tools, and proper safety measures must be followed at all times. If team members have any doubts about using a particular tool or suspect that it may be unsafe, they should seek assistance from experienced team members or mentors. Safety glasses are a mandatory requirement to protect the eyes from potential hazards, while the use of safety gloves depends on the specific tool and associated risks.

Certain tools, such as the deburring tool, soldering iron, hand saw, and angle grinder, require the use of safety gloves to protect team members' hands from potential cuts, burns, or abrasions. On the other hand, tools like the drill press and rotating equipment should not be used with safety gloves, as gloves can pose entanglement hazards and increase the risk of serious injuries.

By adhering to safety guidelines and providing proper safety gear when needed, team members can confidently operate power tools while minimising the risk of injuries. This emphasis on safety creates a culture of responsible and safe practices within the FRC community, ensuring a secure and productive environment for all involved in the robot's construction and assembly.

10.4 Equipment Safety

In addition to power tools and robot safety, it is crucial to be conscious of your safety when handling equipment in FIRST. Equipment can range from batteries to hand tools, and special care should be taken to follow a few important rules:

Proper Handling: Always handle equipment with care and follow the manufacturer's guidelines and instructions. Mishandling equipment can lead to accidents or damage to the equipment itself.

- **Storage and Transportation**: Store and transport equipment appropriately to prevent any damage or potential hazards. Securely store batteries and tools to avoid accidents or mishaps.

- **Battery Safety**: Batteries are a crucial component of the robot and should be handled with care. Follow battery safety guidelines, such as proper charging, storage, and handling, to avoid incidents like overheating or leakage.

- **Tool Safety**: When using hand tools, ensure they are in good working condition and use them only for their intended purposes. Maintain a clean and organised workspace to prevent accidents caused by misplaced tools.

- **Personal Protective Equipment (PPE)**: Depending on the equipment being handled, wearing appropriate PPE such as safety glasses or gloves may be necessary to protect yourself from potential hazards.

By being mindful of these safety rules, team members can reduce the risk of accidents and create a safe environment during all aspects of the FRC activities. Emphasising safety consciousness in equipment handling enhances the overall safety culture within the FIRST community, ensuring that everyone can participate in the program with confidence and security.

10.4.1 Battery Safety

Handling batteries safely is of utmost importance as they contain sulfuric acid (H2SO4), a colourless liquid that can cause severe burns to the eyes, skin, and clothes. If you suspect a battery is damaged or not functioning correctly, immediately remove it from operation to prevent rapid heating and the risk of explosion.

When charging batteries, the location must be well-ventilated to avoid charger failure due to inadequate ventilation. Never touch the battery terminals with a metal device simultaneously, as this can cause the battery to short out and potentially explode. Additionally, ensure you do not charge the battery faster than the manufacturer's maximum recommended rate to prevent any hazards.

FIRST-provided batteries also contain sulfuric acid, and handling them requires caution. If a battery is damaged and acid comes into contact with your skin, follow these steps:

1. Wash the affected area with a large quantity of water.

2. Seek medical treatment promptly.

3. Treat the battery as hazardous material and follow the battery's Safety Data Sheets (SDS).

4. Remove the battery from use and label it as unsuitable for further use.

In the event of an electrolyte leak (leaking battery), follow these measures:

1. Neutralise the acid with sodium bicarbonate (baking soda).
2. Adhere to SDS guidelines.
3. Inform a mentor of the leak.
4. Wear gloves before handling the battery.
5. Place the battery in a leak-proof container for proper disposal.
6. Neutralise the gloves with sodium bicarbonate before storing them.
7. Seek medical help if the skin comes in contact with acid.
8. Dispose of the battery correctly.

Regularly inspect the battery for damage and check it after each round of competition for any signs of potential harm. Following these safety protocols ensures the safe handling and use of batteries, reducing the risk of accidents and promoting a safety-conscious environment within the FRC community.

10.4.2 Electrical Safety

Before using any equipment or extension cord, it is essential to conduct a thorough check for damages to ensure they are in good condition and safe to use. Damaged cords can pose electrical hazards, so it is crucial to address any issues before proceeding with their use.

Avoid daisy-chaining, which involves plugging one power strip into another, as it may cause the circuit to overload and potentially lead to an electrical fire. Prevent overloading by refraining from connecting an extension cord to another extension cord or plugging an extension cord into a power strip. Similarly, avoid plugging a multi-device adapter into a power strip or extension cord.

By adhering to these precautions, team members can significantly reduce the risk of electrical accidents and potential fires. Emphasising safe practices when handling electrical equipment and cords enhances the overall safety culture within the FRC community, promoting a secure working environment for all team members.

10.4.3 SDS

Safety Data Sheets (SDS) are vital documents that provide essential information about hazardous substances and materials used in the FRC environment. SDS contains detailed information about the properties, handling, storage, and emergency procedures related to these substances.

When dealing with hazardous materials, it is crucial to have access to SDS for each substance. Before using or handling any hazardous material, team members should review the corresponding SDS to understand the potential risks and necessary safety precautions.

SDS typically includes the following information:

- Identification: The product name, manufacturer information, and emergency contact details are provided.

- Hazard Identification: This section outlines the potential hazards associated with the substance, including physical, health, and environmental hazards.

- Composition/Ingredients: The SDS lists the ingredients of the material and their respective concentration levels.

- First Aid Measures: Information on first aid procedures to be followed in case of exposure or contact with the hazardous material.

- Firefighting Measures: Instructions on how to handle fires involving the substance and suitable fire-extinguishing methods.
- Accidental Release Measures: Steps to be taken in the event of a spill or release of the material.
- Handling and Storage: Safety guidelines for proper handling, storage, and transportation of the substance.
- Exposure Controls/Personal Protection: Information on protective measures and personal protective equipment (PPE) required when handling the material.
- Physical and Chemical Properties: Details about the physical and chemical characteristics of the substance.
- Stability and Reactivity: Information on the stability of the material and any potential reactions with other substances.
- Toxicological Information: Data related to the toxicological properties of the substance.
- Ecological Information: The impact of the material on the environment.
- Disposal Considerations: Proper methods of disposing of the material.
- Transport Information: Guidelines for the safe transportation of the substance.
- Regulatory Information: Relevant regulatory information and compliance requirements.
- Other Information: Additional relevant information that is not covered in the above sections.

SDS serves as a valuable resource to ensure that team members are well-informed and can handle hazardous materials safely. By following the instructions and guidelines provided in SDS, team members can minimise risks and ensure a safe working environment during all aspects of the FRC activities.

10.4.4 Conclusion

In conclusion, in the FRC community, prioritising safety is crucial when handling batteries and electrical equipment. Batteries containing sulfuric acid pose risks of severe burns and potential explosions if damaged or improperly charged. Following safety protocols, such as prompt removal of damaged batteries and ensuring proper ventilation during charging, is essential to mitigate hazards.

Electrical safety is equally vital, and thorough checks for damages on equipment and cords are necessary to prevent electrical hazards. Avoiding daisy-chaining and overloading circuits with extension cords or power strips significantly reduces the risk of electrical fires. Promoting a strong safety culture within the FRC community fosters a secure working environment for all team members.

Accessing Safety Data Sheets (SDS) for hazardous substances is essential. Reviewing SDS before handling these materials provides critical information on risks and safety precautions. By following the guidelines in SDS, FRC teams can minimise hazards and ensure the safe handling of hazardous materials throughout their activities.

Overall, prioritising safety in battery and electrical equipment handling, as well as adhering to SDS guidelines for hazardous materials, creates a safety-conscious environment within the FRC community, safeguarding the well-being of all team members.

10.5 Hand Tool Safety

Hand Tool Safety is a critical aspect of FRC safety training, emphasising the proper and safe use of hand tools during robot construction and maintenance. Hand tools are commonly used in various tasks, and understanding their safe usage is essential to prevent accidents and injuries.

10.5.1 Safety

When constructing the robot you will need a variety of hand tools. A hand tool is any hand-held tool used to accomplish a task. An important thing to remember is to always use the correct tool for the correct task(example: do NOT use a wrench as a hammer or a screwdriver for a chisel).

When using hand tools, it is essential to follow these safety guidelines:

1. **Selecting the Right Tool**: Choose the appropriate hand tool for the task at hand. Using the wrong tool can lead to damage to the tool or the workpiece and increase the risk of injury.

2. **Inspecting Tools**: Before use, carefully inspect hand tools for any signs of damage, wear, or defects. Damaged tools should not be used and must be replaced or repaired.

3. **Wearing Personal Protective Equipment (PPE)**: Always wear the appropriate PPE, such as safety glasses, gloves, or hearing protection, depending on the tool and the task.

4. **Using the Correct Technique**: Follow proper techniques while using hand tools. Avoid using excessive force or awkward positions that may strain muscles or cause injury.

5. **Maintaining a Clean Workspace**: Keep the work area clean and organised to prevent accidents caused by clutter or tripping hazards.

6. **Using Tools for Their Intended Purpose**: Hand tools should be used only for their designed tasks. Misusing tools can lead to damage, accidents, and injuries.

7. **Carrying Tools Safely**: When transporting hand tools, carry them securely and ensure the sharp edges or points are properly covered.

8. **Properly Storing Tools**: Store hand tools in designated areas, keeping them clean and in good condition. Avoid leaving tools lying around, as they can become tripping hazards.

9. **Handling Sharp Tools with Caution**: Be extremely careful when using sharp hand tools like knives, scissors, or blades. Always cut away from yourself and others.

10. **Working with Others**: Communicate with team members and maintain a safe distance from others when using hand tools to prevent accidental injuries.

By emphasising hand tool safety and ensuring that team members are well-trained in their proper usage, the risk of accidents and injuries during robot construction and maintenance can be significantly reduced. Hand tool safety is a fundamental aspect of FRC safety protocols, fostering a culture of responsible and secure work practices within the team.

10.5.2 Rules for Hand Tools

Tool safety is of utmost importance in FRC, and adhering to proper procedures is crucial to prevent accidents and injuries. Before using any tool, always inspect it to ensure it is in good condition. If you discover a damaged, broken, or defective tool, do not return it to its original location. Instead, remove the tool from service and promptly report the issue to a safety captain or mentor for repair or replacement.

When working on a workpiece, it is essential to place it on a flat, sturdy surface. Avoid holding or carrying the workpiece, as it can lead to unstable working conditions and potential hazards. Always prioritise safety by using a stable work surface.

For hand-held blades, such as knives, wearing protective gloves is essential. Additionally, ensure your knife strokes are directed away from your body to minimise the risk of bodily harm. By following these precautions, you can significantly reduce the chances of injuries related to hand tools.

To summarise the key rules:

1. If a tool is damaged, remove it from use and report it for repair or replacement.
2. Use a flat, sturdy surface when working on a workpiece to ensure stability and safety.
3. When using hand-held blades, wear protective gloves and direct knife strokes away from your body to prevent accidents.

By following these guidelines, FRC team members can create a safe and secure working environment, minimising the risk of injuries and accidents while handling tools.

10.5.3 Storage

Proper storage of tools is essential to maintain a safe workspace and prevent potential hazards. When storing sharp and pointed tools, ensure they are kept in a designated, secure location to avoid accidental injuries. When carrying these tools, always make sure to cover the sharp edge or point to reduce the risk of accidents.

Avoid carrying uncovered tools in your pockets, as it can lead to unintended injuries or damage to the tools. Instead, utilise tool belts, toolboxes, or other appropriate storage methods to keep tools securely during transportation.

Furthermore, never leave tools on an overhang or any precarious surface. Tools left in such positions can fall and pose a risk of injury to individuals below. Always find a stable and safe location for tool storage to minimise the possibility of accidents.

Finally, be mindful of the storage location chosen for equipment. Store tools where they will not cause safety hazards and are protected from potential damage. By ensuring proper storage practices, FRC team members can maintain a safe working environment and protect both themselves and others from avoidable accidents.

10.5.4 Conclusion

In conclusion, hand tool safety is a crucial aspect of FRC safety training, aimed at promoting proper and secure usage of hand tools during robot construction and maintenance. By adhering to the provided safety guidelines, team members can significantly reduce the risk of accidents and injuries while working with hand tools.

Selecting the right tool for the task, inspecting tools for damage, and wearing appropriate personal protective equipment are fundamental practices in hand tool safety. Following correct techniques, maintaining a clean workspace, and using tools for their intended purpose further contribute to a safe working environment. It is essential to handle sharp tools with caution and communicate with team members to prevent accidental injuries.

Adhering to the outlined rules, such as promptly reporting damaged tools and using a stable work surface, is paramount to maintain a secure work environment. Proper storage of tools, especially sharp and pointed ones, ensures a safe workspace and prevents potential hazards.

By emphasising hand tool safety and responsible work practices, FRC teams can foster a culture of safety and prevent avoidable accidents. Prioritising hand tool safety is a fundamental aspect of FRC safety protocols, ensuring a productive and secure working environment for all team members.

11 Tools

In this section, we will explore various hand tools that may be useful during the design and construction of your robot. Some of the tools covered include a power drill, callipers, and a centre punch, among others. However, keep in mind that this list is not exhaustive, and you might encounter new tools throughout the build season.

When using any tool, prioritise safety. If you are unsure about how to use a tool properly, refrain from using it until someone experienced can guide you and ensure your safety. Always seek assistance from mentors, the manufacturing lead, or the safety captain to learn how to use unfamiliar tools safely and effectively.

Throughout the build season, there is a possibility of acquiring new tools not covered in this curriculum. If you come across such tools, do not hesitate to seek guidance from knowledgeable team members to understand their safe usage. Remember, safety should always be the top priority in any manufacturing or building activity.

11.1 Understanding the Use and Safety

Understanding the purpose of each tool and using them as intended is vital for a safe and productive workspace. It is essential never to use a tool for a purpose it was not designed for, as this can lead to accidents and damage the tool itself.

While many of the tools and safety precautions might seem self-explanatory, additional resources such as videos and websites are provided to offer more in-depth guidance. If you encounter a tool you are unsure about, do not hesitate to seek help from a mentor or a fellow team member. Guessing or attempting to use a tool without proper knowledge can result in severe injuries or accidents that could be avoided with proper guidance.

Remember, safety is paramount when working with tools. Always seek assistance and ensure you understand the correct usage of each tool to prevent any potential accidents or harm to yourself or others. Under this there are a couple commonly used tools in robotics that you will encounter at some point while you're manufacturing

11.1.1 Power Drill

The portable power drill is a versatile tool commonly used for drilling holes and driving screws or bolts. It is designed to make tasks like creating openings for fasteners or assembling components much more efficient. When using this tool, it is crucial to prioritise safety and follow the recommended guidelines:

Use:

- Drilling: The power drill is employed to create precise holes in various materials, such as wood, metal, or plastic. These holes serve as anchor points for screws, bolts, or other fasteners.
- Screwdriving: The power drill can be equipped with a screwdriver bit to quickly and effectively drive screws into the material.

Safety while using:

- Wear safety glasses to protect your eyes from flying debris and potential hazards.
- Keep drill bits sharp for efficient drilling and to reduce the risk of accidents.
- Ensure cords are kept away from cutting areas to prevent tripping hazards and potential damage to the cord.
- Check wires for any signs of damage or frays before using the drill to avoid electrical hazards.
- Tighten the chuck securely to ensure drill bits stay in place during operation.
- Always secure the workpiece firmly to prevent movement during drilling or screwing.
- When drilling a large hole, start with a small pivot hole to guide the drill and prevent slipping.
- Avoid using bent drill bits, as they can lead to inaccurate drilling and potential damage.

- Do not reach under or over the workpiece while the drill is in operation to prevent accidental contact with moving parts.

By understanding the proper use and following these safety measures, you can maximise the efficiency and minimise the risk of accidents and injuries while using a portable power drill.

11.1.2 Impact Driver

Impact drivers are powerful tools used for driving screws or bolts with greater force and torque compared to a regular power drill. It is essential to understand their specific use and adhere to safety guidelines to prevent accidents and ensure effective operation:

Use:

- Screwdriving: Impact drivers are designed solely for fastening screws or bolts into various materials, providing higher torque than a standard power drill.

NOT FOR DRILLING: Unlike a power drill, an impact driver is not suitable for drilling holes in materials.
Safety while using:

- Wear safety glasses to protect your eyes from potential hazards and flying debris.

- Check wires for any signs of damage or frays before using the impact driver to avoid electrical risks.

- Tighten the chuck securely to ensure the bit remains firmly in place during operation.

- Always secure the workpiece firmly before screwdriving to prevent movement.

- When screwing in a bolt, it is advisable to drill a small pivot hole first to guide the fastener accurately.

- Avoid reaching under or over the workpiece while using the impact driver to prevent accidental contact with moving parts.

- Use the correct chuck for the specific bit, as there may be multiple types of the same bit that require different chucks.

TIP: When using an impact driver to remove screws, press towards the direction the screw is facing to allow the threads to have a better grip on the material, making the removal process more effective.

By adhering to these guidelines and understanding the purpose of an impact driver, you can work safely and efficiently when fastening screws or bolts without the risk of damage or accidents.

11.1.3 Handsaws

Use:

- Hand saws are used to cut through wood into small, precise pieces.

Safety while using:

- Check the saw blade for any damage or defects before use to ensure safe cutting.

- Wear safety glasses to protect your eyes from potential flying wood particles or debris.

- Start cutting carefully and slowly to prevent the saw blade from jumping or binding during the cut.

- Apply pressure on downward strokes only, avoiding any lateral or sideways pressure on the blade.

- Hold the wood stock firmly in place with clamps or a stable surface to ensure stability during cutting.

- Keep the saw blade clean and free from sawdust or debris buildup for better cutting performance.

- Do not twist the saw or the wood when applying pressure during the cut to maintain control and accuracy.

By adhering to these safety guidelines, you can use hand saws effectively and reduce the risk of accidents or injuries while cutting wood. Remember to prioritise safety at all times, and if you are unsure about using the tool, seek guidance from a mentor or an experienced team member to ensure proper and safe operation.

11.1.4 Screwdrivers

Use:

- Screwdrivers are used to screw in screws.

Safety while using:

- Use the correct screwdriver for the type and size of screw you are working with to ensure a proper fit and prevent damage to the screw head.

- Wear safety glasses to protect your eyes from any potential flying particles or debris while using the screwdriver.

- If a conventional screwdriver cannot be safely used due to limited access, use an offset screwdriver for better reach and control.

- Do not apply excessive force or push against the screwdriver with more force than necessary to keep the screw in place; this may lead to slipping or damaging the screw head.

- Avoid holding the workpiece with one hand while using the screwdriver with the other hand; instead, secure the workpiece using clamps or a stable surface.

- Do not use defective or broken screwdrivers, as they may lead to slipping or breaking during use, increasing the risk of injury.

By following these safety measures, you can use screwdrivers safely and efficiently. Always prioritise safety and use the right tool for the job to minimise the risk of accidents and ensure successful screwing tasks. If you encounter any uncertainties or challenges, seek guidance from a mentor or experienced team member to ensure safe operation.

11.1.5 Snips

Use:

- Snips are used to cut sheet metal or other soft materials.

Safety while using:

- Select the correct type of snips for the specific cutting job you need to perform. Choose between straight, wide curve, tight curve, right-cut, or left-cut snips based on the cutting requirements.

- Only use snips for cutting soft metals; avoid using them on harder materials as it may damage the snips and result in an ineffective cut.

- Engage the locking clip when the snips are not in use to prevent accidental opening and potential injury.

- Avoid trying to cut sharp curves using straight snips, as they are not designed for such tasks and may cause damage to the snips or result in an inaccurate cut.

- Do not use a hammer or your feet to apply additional pressure to the snips, as this may lead to damage or affect the precision of the cut.

- Never attempt to resharpen snips using a file, as this can alter the shape and functionality of the cutting edges, reducing the snips' effectiveness.

By following these safety guidelines, you can use snips effectively and reduce the risk of accidents while cutting sheet metal or other soft materials. Always choose the appropriate type of snips for the task and handle them with care to ensure both safety and accurate cuts. If you have any uncertainties or concerns about using snips, seek advice from a mentor or experienced team member to ensure safe and efficient operation.

11.1.6 Pliers and Wire Cutters

Use:

- Side Cutting Pliers: Used for various purposes, including electrical and manufacturing tasks.

- Long Nose Pliers: Used to grip small objects, reach into awkward places, and hold or attach wires. Utility Pliers: Used to grip flat, square, or hexagonal pieces. Flat-Nose Pliers: Employed in many applications to grip, turn, and bend wires.

Safety while using:

- Wear safety glasses to protect your eyes from the risk of flying wires or small objects.
- Cut wires at the correct angles to ensure clean and precise cuts.
- Use pliers in good condition, free from any damage or defects.
- Check that the cutting edges of the pliers are sharp to maintain their effectiveness.
- Choose pliers that allow you to hold the workpiece comfortably and securely to prevent slipping or mishandling.

By adhering to these safety practices, you can effectively and safely use various types of pliers for different tasks. Always ensure your pliers are in good working condition and use them at the appropriate angles to achieve accurate and secure grip and cuts. If you encounter any issues or uncertainties with using pliers, seek guidance from mentors or experienced team members to ensure proper and safe usage.

11.1.7 Mire Saw/Cop Saw

Use:

- Power saws are used to cut things into pieces efficiently and quickly.

Safety while using:

- Always wear safety glasses to protect your eyes from potential flying debris.
- Before using the power saw, check the blade for any damage or defects.
- Ensure the blade is securely attached to the saw before operating it.

- Keep cords away from the cutting area to prevent any accidents or entanglements.
- Examine the wires for any damage or frays to ensure safe operation.
- Securely hold the workpiece in place to prevent it from moving during cutting.
- Make sure there is no one standing directly behind the saw to avoid injury from kickbacks.
- Tie back long hair to prevent it from getting caught in the saw's moving parts.
- Avoid wearing loose clothing that could potentially get entangled with the saw.

By following these safety guidelines, you can use power saws effectively and reduce the risk of accidents or injuries. Prioritise safety at all times and be attentive to potential hazards in your work environment when operating power saws.

11.1.8 Drill Press

Use:

- A drill press is used to drill more precise holes with increased force and accuracy.

Safety while using:

- Always wear safety glasses to protect your eyes from any potential debris.
- Keep the drill bits sharp to ensure efficient drilling and reduce the risk of accidents.
- Ensure cords are kept away from the cutting areas to avoid any entanglement hazards.

- Before use, check the wires for any damage or frays to ensure safe operation.
- Tighten the chuck securely to prevent the drill bit from slipping during drilling.
- Securely hold the workpiece in place to prevent it from moving while drilling.
- Drill a small pivot hole before drilling a larger one to help maintain precision.
- Do not use bent drill pieces, as they may not drill accurately and can be hazardous.
- Avoid reaching under or over the workpiece while the drill press is in operation.

Following these safety precautions when using a drill press will help ensure a safe and effective drilling process. Always prioritise safety in your workshop and use the drill press with caution and attention to potential risks.

11.1.9 Rivet Gun

Use:

- A riveter is used to secure two sheets of metal together by joining them with rivets.

Safety while using:

- Always wear safety glasses to protect your eyes from potential debris or metal fragments.
- Ensure the workpiece is securely held in place to prevent any movement during riveting.
- Drill holes in the metal sheets to the appropriate size before using the riveter.

- Avoid using bent or broken rivets, as they may not provide a secure and reliable connection.

By following these safety measures when using a riveter, you can minimise the risk of accidents and ensure a successful and safe metal joining process. Always prioritise safety and use the riveter responsibly and effectively.

11.1.10 Calliper

Use:

- Callipers are used to precisely measure the dimensions of various objects.

Safety while using:

- Before using the calliper, ensure your surroundings are safe and free from any potential hazards.
- If others are working nearby, wear eye protection to protect yourself from any flying debris.

Tips:

- There are multiple ways to measure using a calliper, depending on the specific dimensions you need to determine.
- The main method involves measuring the distance between the two "claws" of the calliper.
- The top part of the calliper can be used to measure the inside diameter of a tube or pipe.
- The protruding piece at the bottom can be utilised to measure depth.

By following safety precautions and understanding the various measurement techniques, you can make accurate and safe use of the calliper for precise measurements. Always be cautious and attentive while handling this tool to prevent accidents and ensure reliable measurements.

11.1.11 Vice

Use:

- A vise is used to securely hold workpieces in place during various tasks.

Safety while using:

- Depending on your surroundings and the nature of the work, wear safety glasses to protect your eyes.
- Keep your hands and other body parts outside of the vise while it is in operation.
- Be mindful of the workpieces to ensure they do not get damaged or crushed during clamping.

Tips:

- Exercise caution not to overtighten the vise, as excessive force may damage delicate workpieces, especially extrusions.
- When using the vise, apply firm but controlled pressure to securely hold the workpiece without causing any harm.

By following these safety guidelines and using the vise appropriately, you can work effectively and safely with various materials and projects. Always be attentive and cautious when using the vise to prevent accidents and protect your workpieces.

11.1.12 Chain Tool

Use:

- A chain breaker is a tool used to separate or break apart chain links during maintenance or repair tasks.

Safety while using:

- Wear safety glasses to protect your eyes from any flying debris or metal fragments that may result from using the chain breaker.

- Ensure the workpiece (the chain) is securely positioned within the chain breaker before applying pressure to break it apart.

By taking these safety precautions, you can safely and effectively use the chain breaker to work with chains, minimising the risk of injuries and accidents. Always prioritise safety when using tools to maintain a safe working environment.

11.1.13 Files

Use:

- A file is a tool used to smooth out rough edges and remove burrs from workpieces.

Safety while using:

- Wear safety glasses to protect your eyes from any debris or metal shavings that may be generated while filing.

- When filing, always work from top to bottom in a controlled manner to avoid accidentally cutting yourself or damaging the workpiece.
- Secure the workpiece properly to prevent any movement or slipping during filing.

Tips:

- Avoid over-filing a piece, as it may result in it becoming too short or losing its intended shape. Work carefully and check your progress regularly to achieve the desired outcome without overdoing it

.

By following these safety guidelines and tips, you can use a file effectively and safely for various projects and tasks. Safety should always be a top priority when working with tools to prevent accidents and injuries.

11.1.14 Deburring Tool

Use:

- A deburring tool is employed to remove burrs and smooth out the edges of drilled holes and workpiece surfaces.

Safety while using:

- Wear safety glasses to protect your eyes from any flying debris or metal fragments.
- Ensure that the blade of the deburring tool is sharp, as a dull blade may not work efficiently and could be hazardous.

- Wear gloves while using the deburring tool to safeguard your hands from potential cuts or injuries.

- Properly secure the workpiece to prevent movement or slipping during the deburring process.

Tips:

- Apply the right amount of force while deburring to effectively remove burrs and achieve a smooth surface.

- Avoid over-deburring a hole, as it may result in it becoming too large for rivets or other fasteners.

By adhering to these safety precautions and tips, you can use a deburring tool effectively and safely to achieve clean and polished edges on your workpieces. Always prioritise safety when using any tool to minimise the risk of accidents and ensure a successful outcome in your projects.

11.1.15 Soldering Iron

Use:

- A soldering iron is used to join wires together using metal

Safety while using:

- Wear safety glasses to protect your eyes from potential splatters of hot solder or other hazards.

- Use gloves to shield your hands from accidental burns while handling the soldering iron.

- Employ a heat sink or stand to place the soldering iron when not in use, preventing accidental contact with surfaces and reducing the risk of burns.

- Strip wires properly and ensure they are clean before soldering to achieve a reliable connection.

- Work in a well-ventilated area to disperse soldering fumes and prevent inhalation of harmful substances.

- Wear a mask to avoid breathing in toxic and potentially carcinogenic solder fumes.

Tips:

- Be cautious of solder fumes, as they can be harmful to your health. Proper ventilation and wearing a mask are essential to protect yourself.

- Ensure that the soldering iron's temperature is set appropriately for the task at hand to achieve proper solder flow and prevent overheating.

- Remember to "tin" the tip of the soldering iron before soldering, which involves applying a small amount of solder to the tip for better heat transfer and performance.

- Have the heat shrink tubing prepared before soldering to insulate and protect the soldered connection.

By adhering to these safety precautions and tips, you can safely and effectively use a soldering iron for your wiring needs. Prioritising safety is crucial to prevent accidents and potential health risks associated with soldering operations.

11.1.16 Wire Strippers

Use:

- Wire strippers are employed to remove the sheathing from wires or to strip the insulation off individual wires.

Safety while using:

- Ensure that the wires are disconnected from any power source before using the wire strippers to avoid electrical shock or short circuits.

- Wear safety glasses to protect your eyes from potential flying debris or sharp edges.

Tips:

- Use the appropriate wire gauge slot on the wire strippers to match the size of the wire you are stripping. If you are unsure about the wire gauge, don't hesitate to ask a mentor or experienced team member for assistance.

By following these safety guidelines and tips, you can safely and accurately use wire strippers for your electrical work. Always prioritise safety to prevent accidents and ensure a successful wire-stripping process.

11.2 Tool Conclusion

In conclusion, maintaining a safe and productive workspace is of utmost importance when working with tools. Understanding the purpose of each tool and using them as intended is essential to prevent accidents and injuries. It is crucial to prioritise safety and never use a tool for tasks it was not designed for, as this can lead to damage to the tool and potential harm to oneself or others.

The provided guidelines and safety tips for various tools, such as portable power drills, impact drivers, hand saws, screwdrivers, snips, pliers and wire cutters, mitre saws, drill presses, rivet guns, callipers, vises, chain tools, files, deburring tools, soldering irons, and wire strippers, serve as valuable resources for ensuring safe and effective usage.

Remember to wear appropriate safety gear, such as safety glasses and gloves, when working with tools. Seek help from mentors or experienced team members if uncertain about a tool's proper use to avoid accidents and injuries. By adhering to safety measures and using the tools responsibly, one can create a secure and efficient work environment while achieving successful outcomes in various projects. Prioritising safety is key to fostering a culture of carefulness, caution, and productivity in the workspace.

12 Hardware

There is a wide array of materials available for building the robot, each with its unique properties, strengths, and advantages. When selecting materials, considerations like strength, durability, weight, and other factors play a crucial role in determining the overall performance of the robot. In FRC, one commonly used component is the nyloc nut, especially in high-frequency environments, which the robot falls under.

By considering the properties and applications of various materials and components like the nyloc nuts, you can make informed decisions to build a robust and efficient robot for FRC competitions. Always prioritise safety when handling materials to prevent any potential accidents or injuries during the construction process.

12.1 Aluminium Tubing

In our team, we prioritise using 6061-T6 aluminium for manufacturing various robot parts. Understanding the naming conventions for aluminium alloys is crucial to making informed material choices.

Advantages of 6061 Aluminium

- 6061-T6 aluminium is known for its excellent strength-to-weight ratio, making it suitable for various structural components of the robot.

- It offers good machinability, making it easier to work with and fabricate precise parts.
- This aluminium alloy exhibits good corrosion resistance, ensuring the longevity of robot components.

Safety Precations
When working with metal alloys, always wear appropriate personal protective equipment (PPE) like safety glasses and gloves to safeguard against sharp edges and potential hazards.

By comprehending the aluminium alloy naming conventions and considering the properties of 6061 alloy, our team makes informed material choices and employs safety measures to build a robust and reliable robot for FRC competitions.

12.2 Naming Conventions

In this section we will learn about naming aluminium, the different types of aluminium, and other metals.

12.2.1 Composition

The naming conventions for aluminium alloys provide valuable information about their composition and treatments. Understanding these codes helps us make informed decisions when selecting materials for our robot.

The first digit signifies the primary alloying element:

- 1xx: Minimum 99.000% Aluminium
- 2xx: Copper

- 3xx: Manganese
- 4xx: Silicon
- 5xx: Magnesium
- 6xx: Magnesium and Silicon
- 7xx: Zinc
- 8xx: Other Elements

The second digit indicates a standard alloy with no alteration in composition.

The last two digits, "61," represent a specific alloy composition. In this case, referring to the chart, the alloy "6061" contains 1% magnesium and 0.6% silicon.

		Composition of aluminum alloys			
Alloy	%Cu	%Mg	%Mn	%Si	%Zn
2024	4.4	1.5	0.6	0	0
6061	0	1	0	0.6	0
7005	0	1.4	0	0	4.5
7075	1.6	2.5	0	0	5.6
356.0	0	0.3	0	7	0

Figure 11: Composition Table with Name
[**davisalum**]

12.2.2 Further Treatment

Following the four-digit code, there is a letter and additional numbers that provide details about the metal's treatments, such as hardening and tempers.

For example, "T6" indicates that the alloy has undergone an artificial ageing process after cooling from a solution heat treatment.

After the 4 digits there is a letter and some numbers that tells us what treatments(hardening, tempers...) the metal has been through. The T6 tells us that the alloy has been artificially aged after cooling from a solution heat treatment.

12.2.3 Reference Material

This video explains when and why we use 6061-T6 aluminium alloy and 5052-H32 aluminium alloy for the robot

Rapid Sheet Metal® - Rapid Tech Tip: 6061-T6 vs. 5052-H32

This websites goes over the different tempers the alloy can go through

Temper Designations for Aluminium - Metal Supermarkets

12.3 Conclusion

Our team prioritises using 6061-T6 aluminium due to its excellent strength-to-weight ratio, machinability, and corrosion resistance for manufacturing various robot parts. Understanding the naming conventions for aluminium alloys helps us make informed material choices. We emphasise safety by wearing appropriate protective equipment when working with metal alloys. This knowledge enables us to build robust and reliable robots for FRC competitions, ensuring optimal performance and longevity.

13 Motors

In FRC, selecting the right motor for each scenario is crucial to maximise efficiency while being mindful of our budget constraints. The two main types of motors used in FRC are brushed and brushless motors, each offering distinct advantages and trade-offs.

Brushed Motors

- Brushed motors are generally less efficient, emit more noise, and wear out more quickly compared to brushless motors.
- However, they are more cost-effective and can typically handle heavier loads, making them suitable for certain applications.
- Due to their affordability, brushed motors are commonly used in scenarios where budget is a concern.

Brushless Motors

- Brushless motors outperform brushed motors in almost every aspect.
- They are more efficient, quieter, and have longer lifespans, providing a smoother and more reliable operation.
- Despite their superior performance, brushless motors come with increased complexity in terms of control and operation.
- Their higher cost can also be a limiting factor, especially for teams with limited financial resources.

Key Considerations

- **Efficiency**: Choose motors based on the desired level of efficiency in different robot subsystems. High-efficiency motors can lead to improved overall performance and longer battery life.

- **Noise**: Consider the noise factor when selecting motors, especially in applications where quiet operation is essential for strategic gameplay or team communication.

- **Load Capacity**: Assess the loads and stresses that motors will experience in different robot mechanisms. Opt for motors that can handle the required loads without strain.

- **Budget**: Being mindful of the team's budget, striking a balance between performance and cost is vital. Make informed decisions to optimise the robot's performance while staying within financial constraints.

- **Control Complexity**: Evaluate the team's expertise and familiarity with motor control systems, as brushless motors may require more sophisticated control methods.

By carefully weighing the advantages and disadvantages of brushed and brushless motors, our team can make informed decisions to create an efficient, functional, and cost-effective robot for FRC competitions.

13.1 CIMs

Motor controllers play a crucial role in controlling the operation of motors in a robotics system. In this case, the motor controllers mentioned are the Victor SPX and Victor SPX, which are commonly used in FRC (FIRST Robotics Competition) teams to control CIMS motors.

Pros of using CIMS motors with Victor SPX or Victor SPX motor controllers:

- **Strength and Load Capacity**: CIMS motors are known for their robustness and ability to handle heavy loads. They are well-suited for powering mechanisms that require significant force, such as drivetrains and lifting mechanisms.

- **Torque**: CIMS motors provide a decent amount of torque, which is essential for tasks that demand strong rotational force, like moving heavy game pieces or overcoming resistance in certain mechanisms.

- **User-Friendly**: CIMS motors are brushed motors, which means they are relatively straightforward to use and control. Teams with varying levels of experience can easily integrate and operate these motors into their robot designs.

- **Cost-Effectiveness**: CIMS motors are generally more affordable compared to some of their more advanced counterparts, making them a popular choice for FRC teams with budget constraints.

Cons of using CIMS motors with Victor SPX or Victor SPX motor controllers:

- **Weight**: One notable drawback of CIMS motors is their weight. Their substantial build can add to the overall weight of the robot, potentially affecting agility and manoeuvrability.

- **Efficiency**: While CIMS motors offer excellent strength and load capacity, they are not as efficient as some brushless motors. This means they may consume more power and generate more heat during operation, impacting the overall efficiency of the robot.

- **Speed and Power**: While CIMS motors provide sufficient torque, they may not be the fastest or most powerful motors available in the market. This limitation might affect the robot's top speed and its ability to perform tasks requiring high-speed motion.

In conclusion, using CIMS motors with Victor SPX or Victor SPX motor controllers in an FRC robot design comes with distinct advantages and trade-offs. These motors excel in providing strength and torque, making them suitable for applications requiring substantial force. Additionally, their user-friendly nature and cost-effectiveness make them a popular choice for many FRC teams. However, teams must also consider factors like weight, efficiency, and speed while incorporating CIMS motors into their robot designs to ensure optimal performance on the field.

13.2 Mini CIMs

The motor controllers mentioned, the Victor SPX or Victor SPX, are commonly used in robotics, particularly in FRC (FIRST Robotics Competition) teams, to control Mini CIMS motors. Let's delve into the pros and cons of using Mini CIMS motors with these motor controllers:

Pros of using Mini CIMS motors with Victor SPX or Victor SPX motor controllers:

- **Strength and Load Capacity**: Mini CIMS motors are known for their strength and ability to handle heavy loads. Despite their smaller size compared to standard CIMS motors, they are still powerful and suitable for applications that require significant force, such as driving mechanisms and lifting mechanisms.

- **Torque**: Mini CIMS motors provide a decent amount of torque, making them well-suited for tasks that demand strong rotational force. This characteristic allows them to power mechanisms requiring substantial pulling or pushing force.

- **User-Friendly**: Similar to standard CIMS motors, Mini CIMS are brushed motors, which means they are relatively straightforward to use and control. This user-friendly nature allows teams with varying levels of experience to easily integrate and operate these motors in their robot designs.

- **Cost-Effective**: Mini CIMS motors are generally more affordable compared to more advanced and specialised motors. This cost-effectiveness is advantageous for FRC teams with budget constraints, as they can achieve decent performance without overspending on motors.

Cons of using Mini CIMS motors with Victor SPX or Victor SPX motor controllers:

- **Weight**: Although Mini CIMS motors are smaller than standard CIMS motors, they can still add a notable amount of weight to the robot. This added weight might affect the robot's agility and manoeuvrability, especially if multiple Mini CIMS are used throughout the design.

- **Efficiency**: While Mini CIMS motors offer considerable strength and load capacity, they are not as efficient as some brushless motors. This means they may consume more power and generate more heat during operation, potentially impacting the overall efficiency of the robot.

- **Speed and Power**: Although Mini CIMS motors are strong, they might not be the fastest or most powerful motors available in the market. Teams should consider this limitation when selecting motors for tasks requiring high-speed motion or maximum pushing force.

In summary, using Mini CIMS motors with Victor SPX or Victor SPX motor controllers presents a combination of advantages and trade-offs. These motors offer considerable strength, torque, and user-friendliness, making them a popular choice for various mechanisms in FRC robots. Their affordability is particularly beneficial for teams on a budget. However, teams should also consider factors such as weight, efficiency, and desired speed when incorporating Mini CIMS motors into their robot designs to ensure optimal performance and competitiveness in FRC competitions.

13.3 775 Pro/775 RedLine

Using Talon SRX or Victor SPX motor controllers with 775 Pro or 775 RedLine motors offers specific advantages and drawbacks that teams should consider when designing their FRC robots.

Pros of using 775 Pro or 775 RedLine motors with Talon SRX or Victor SPX motor controllers:

- **Speed**: 775 Pro and 775 RedLine motors are known for their high-speed capabilities, making them some of the fastest motors available for FRC robots. Their quick rotational speed is valuable for applications that require rapid motion, such as shooter mechanisms or flywheels.

- **Weight**: Both 775 Pro and 775 RedLine motors are relatively lightweight compared to other motors with similar power outputs. This reduced weight can be advantageous in ensuring a lighter overall robot design, contributing to better agility and manoeuvrability.

- **User-Friendly**: Like other brushed motors, 775 Pro and 775 RedLine motors are straightforward to use and control. This user-friendliness simplifies their integration into the robot's electrical system and facilitates smooth operation.

- **Cost-Effective**: These motors are relatively affordable, which is beneficial for FRC teams with budget limitations. Their cost-effectiveness allows teams to allocate resources to other critical aspects of their robot design without compromising on performance.

- **Efficiency**: 775 Pro and 775 RedLine motors are decently efficient in converting electrical power into mechanical output. This efficiency is beneficial for maximising the robot's battery life and reducing heat generation during prolonged use.

Cons of using 775 Pro or 775 RedLine motors with Talon SRX or Victor SPX motor controllers:

- **Low Torque**: One of the limitations of 775 Pro and 775 RedLine motors is their lower torque compared to other motors like the Falcon motors. This means they might not provide as much force for tasks that demand high pushing or pulling power, such as lifting heavy game pieces.

- **Load Capacity**: These motors might not be suitable for applications that require handling heavy loads or supporting mechanisms that involve significant weight. Their limited load capacity could lead to motor strain or reduced performance in certain scenarios.

In summary, using Talon SRX or Victor SPX motor controllers with 775 Pro or 775 RedLine motors offers a balance of advantages and trade-offs. The motors' high speed, lightweight nature, affordability, and decent efficiency make them valuable options for specific robot mechanisms. However, teams should be mindful of their lower torque and load capacity when selecting these motors for tasks that require more force or involve heavier loads. By carefully considering the robot's requirements and the strengths of these motors, teams can make informed decisions to optimise the robot's performance in FRC competitions.

13.4 Neo 550

Using Spark MAX motor controllers with Neo 550 motors offers specific advantages and considerations that FRC teams should take into account when designing their robots.

Pros of using Neo 550 motors with Spark MAX motor controllers:

- **Speed**: Neo 550 motors are known for their fast rotational speed, making them suitable for applications that require quick motion. Their high speed capabilities are valuable for mechanisms like shooters, conveyors, or other systems requiring rapid movement.

- **Efficiency**: Neo 550 motors are more efficient compared to many other motors commonly used in FRC. Their improved efficiency results in less power loss as heat, maximising the mechanical output for a given electrical input.

- **Compact and Lightweight**: Neo 550 motors are among the smallest and lightest motors available in FRC. Their compact size and reduced weight allow for more flexible and space-efficient robot designs. These motors are particularly useful in situations where weight and space constraints are critical.

Cons of using Neo 550 motors with Spark MAX motor controllers:

- **Cost**: While the Neo 550 motors themselves are relatively affordable, the Spark MAX motor controllers can be relatively expensive compared to other motor controller options. This cost factor should be considered when budgeting for the overall robot design.

- **Speed and Strength**: While Neo 550 motors are fast, they might not match the top-tier speed or pushing power of other specialised motors like the Falcon motors. Teams should carefully evaluate the required speed and torque of their robot mechanisms to ensure Neo 550 motors meet their performance needs.

- **Brushless Complexity**: Neo 550 motors are brushless motors, which can be more complex to control and program compared to brushed motors. Teams should ensure they have the expertise and resources to handle the nuances of brushless motor control to fully leverage the advantages of the Neo 550 motors.

In summary, using Spark MAX motor controllers with Neo 550 motors provides FRC teams with compact, lightweight, and efficient motor options. The motors' high speed and efficiency make them suitable for certain robot mechanisms that demand rapid motion and optimised power usage. However, teams should be mindful of the costs associated with motor controllers and evaluate whether the Neo 550 motors' speed and torque capabilities align with their specific robot design requirements. Additionally, teams should consider their familiarity with brushless motor control and programming to ensure they can effectively utilise the Neo 550 motors' full potential. By carefully weighing the pros and cons, teams can make informed decisions to create a high-performing and efficient robot for FRC competitions.

13.5 Neo

Using Spark MAX motor controllers with Neo motors offers specific advantages and considerations that FRC teams should take into account when designing their robots.

Pros of using Neo motors with Spark MAX motor controllers:

- **Speed**: Neo motors are known for their fast rotational speed, making them suitable for applications that require quick motion. Their high speed capabilities are valuable for mechanisms like shooters, conveyors, or other systems requiring rapid movement.

- **Efficiency**: Neo motors are more efficient compared to many other motors commonly used in FRC. Their improved efficiency results in less power loss as heat, maximising the mechanical output for a given electrical input. This efficiency can lead to longer battery life and overall improved performance.

- **Torque**: Neo motors offer a substantial amount of torque, making them capable of handling various robot mechanisms that require significant pushing power or heavy load-bearing capacities.

- **Brushless Technology**: Neo motors are a brushless motor design, providing several advantages over brushed motors, such as longer lifespan, reduced maintenance, and better performance in high-frequency environments.

- **Compact and Lightweight**: Like brushed Neo 550 motors, Neo motors are relatively compact and lightweight, allowing for more flexible and space-efficient robot designs.

Cons of using Neo motors with Spark MAX motor controllers:

- **Cost**: One of the significant drawbacks of using Neo motors is their higher cost compared to some other motor options. This includes both the cost of the motor itself and the accompanying motor controllers. Teams should carefully consider their budget constraints when choosing Neo motors for their robot.

- **Speed and Strength**: While Neo motors offer impressive speed and torque, they may not match the absolute top-tier speed and pushing power of certain specialised motors like the Falcon motors. Teams should evaluate their specific robot design requirements to ensure that Neo motors meet their performance needs.

- **Brushless Complexity**: Neo motors require brushless motor controllers like Spark MAX, which can be more complex to control and program compared to brushed motors and their corresponding controllers. Teams should ensure they have the expertise and resources to handle the nuances of brushless motor control to fully leverage the advantages of Neo motors.

In summary, using Spark MAX motor controllers with Neo motors provides FRC teams with high-speed, efficient, and powerful motor options. The motors' combination of speed, torque, and efficiency makes them suitable for various robot mechanisms, particularly those requiring quick and robust motion. However, teams should carefully consider the cost implications and evaluate whether Neo motors meet their specific robot performance requirements. Additionally, teams should be prepared to handle the complexities of brushless motor control to effectively utilise the advantages of Neo motors. By thoroughly assessing the pros and cons, teams can make informed decisions to create a high-performing and efficient robot for FRC competitions.

13.6 Falcon 500

Using the Talon SRX motor controller with Falcon 500 motors offers both significant advantages and potential concerns that FRC teams should carefully consider when selecting their robot's motors.

Pros of using Falcon 500 motors with Talon SRX motor controllers:

- **Speed**: Falcon 500 motors are known for their impressive speed capabilities, making them well-suited for mechanisms that require quick and precise motion. Their high speed can be advantageous for various game-specific tasks like shooting game pieces or rapid movement in autonomous modes.

- **Torque**: Falcon 500 motors offer high torque output, providing substantial pushing power and load-carrying capacity. Their strong torque makes them suitable for mechanisms that require high force, such as drive trains, lifting arms, or manipulators.

- **Efficiency**: Falcon 500 motors are designed to be highly efficient, converting electrical power into mechanical output with minimal energy loss as heat. Their efficiency contributes to longer battery life, reduced heating during prolonged use, and overall improved robot performance.

- **Superior Specs**: Falcon 500 motors are engineered with advanced features, making them stand out as one of the top motor options in terms of specifications. Their combination of speed, torque, and efficiency makes them a popular choice for high-performing FRC robots.

Cons of using Falcon 500 motors with Talon SRX motor controllers:

- **Cost**: One of the significant drawbacks of Falcon 500 motors is their higher cost compared to other motor options available in FRC. This includes both the cost of the motor itself and the specific motor controllers required for their operation. The higher price can be a limiting factor for teams with budget constraints.

- **Quality Control**: Over the years, there have been concerns about the quality control of Falcon 500 motors. While they are known for their exceptional performance, some teams have experienced issues with motor reliability, including potential failures or inconsistencies. Teams should be aware of this historical concern and may consider having backup motor options in case of unexpected issues.

In summary, using Talon SRX motor controllers with Falcon 500 motors offers FRC teams a high-speed, high-torque, and efficient motor option. Falcon 500 motors excel in performance, making them an attractive choice for many robot applications. However, teams should carefully consider the higher cost of these motors and be prepared to manage potential quality control concerns. Overall, Falcon 500 motors, when paired with Talon SRX motor controllers, can contribute to the creation of a powerful and capable robot, especially for tasks that require speed, strength, and efficiency.

13.7 Conclusion

In conclusion, selecting the right motor for an FRC robot is vital to maximise efficiency and stay within budget constraints. Brushed motors like Cims and Mini Cims are cost-effective and handle heavy loads, while brushless motors like Neo 550s and Falcons offer higher efficiency and speed. Careful consideration of the motor's specifications and the team's expertise in motor control will lead to a successful and competitive robot design.

14 Electronics

Electronics play a crucial role in powering and controlling the robot in FRC competitions. Having a solid understanding of electronics is essential for the successful operation of the robot, as it directly influences its functionality and performance. Additionally, proper knowledge of electronics is vital for ensuring the safety of both the robot and the team members working on it.

14.1 Powering the Robot

Electronics, such as motor controllers, power distribution panels, and batteries, are responsible for providing the necessary electrical power to drive the robot's various mechanisms. Understanding how to properly connect and configure these components is essential to ensure that the robot functions as intended. Without a good knowledge of electronics, teams may encounter issues with power distribution, motor control, or other critical functions, leading to robot malfunctions and reduced performance.

14.2 Control and Automation

Electronics also facilitate robot control and automation through sensors, motor controllers, and programmable microcontrollers like the RoboRIO. These components enable teams to implement autonomous routines and teleoperated control to perform specific tasks during matches. Understanding the principles of programming and wiring these electronics is essential for the successful execution of game strategies and achieving high scores in competitions.

14.3 Safety Considerations

Knowledge of electronics is not only essential for functionality but also for safety. Mishandling or improper wiring of electrical components can lead to short circuits, electrical hazards, or even fires. Teams must be well-versed in the correct wiring techniques, proper insulation, and safety practices to avoid potential risks during robot operation. This includes using appropriate wire gauges, securely fastening connections, and using appropriate fuses or circuit breakers to prevent electrical overloads.

14.4 Troubleshooting

Throughout the build season and competitions, teams may encounter electrical issues that require quick diagnosis and resolution. Understanding electronics allows team members to identify problems, such as loose connections, faulty components, or programming errors, and efficiently troubleshoot and rectify them. Proper knowledge of electronics empowers teams to react swiftly during competitions, minimising downtime and maximising robot performance.

14.5 Compliance and Inspection

FRC robots must undergo rigorous inspections to ensure they meet safety and technical standards set by the competition rules. Knowledge of electronics is crucial during inspection processes, where inspectors will check the robot's electrical systems for compliance. Familiarity with electronics helps teams pass inspections smoothly and avoid penalties or disqualifications due to non-compliant wiring or unsafe electrical practices.

15 Key Safety and Care

To ensure the safety and reliability of the robot's electrical components, teams must adhere to the following guidelines:

1. Battery Safety (Refer to Section 4.4.1): Properly handle, charge, and store the robot battery following the manufacturer's instructions to prevent acid leakage, electrical shock, or fire

hazards.

2. Electricity Safety (Refer to Section 4.4.2): Implement safe wiring practices, use appropriate wire gauges, and secure connections to prevent electrical hazards and ensure efficient power distribution.

3. Power Distribution Panel and Voltage Regulator Module: Inspect and secure connections to avoid power issues and potential electrical hazards.

4. RoboRIO, Radio, and Robot Signal Light: Handle these components with care, ensuring proper connections and secure mounting to prevent electrical issues during matches.

5. Circuit Breakers: Use the correct rating for each circuit, and avoid bypassing or modifying circuit breakers to ensure safety and protect electrical components.

6. Soldering: Practise proper soldering techniques to create reliable connections and avoid issues with loose or cold joints.

7. CAN Wire and Connectors (Wago and Weidmuller): Inspect and secure CAN connections to ensure reliable data communication between components.

15.1 Conclusion

In conclusion, electronics are the backbone of an FRC robot, and a solid understanding of electronics is indispensable for success in the competition. From powering and controlling the robot to ensuring safety and compliance, electronics knowledge plays a pivotal role in building a functional, efficient, and competitive robot. Teams should invest time in learning and mastering electronics to unleash the full potential of their robot and enhance their overall performance on the field.